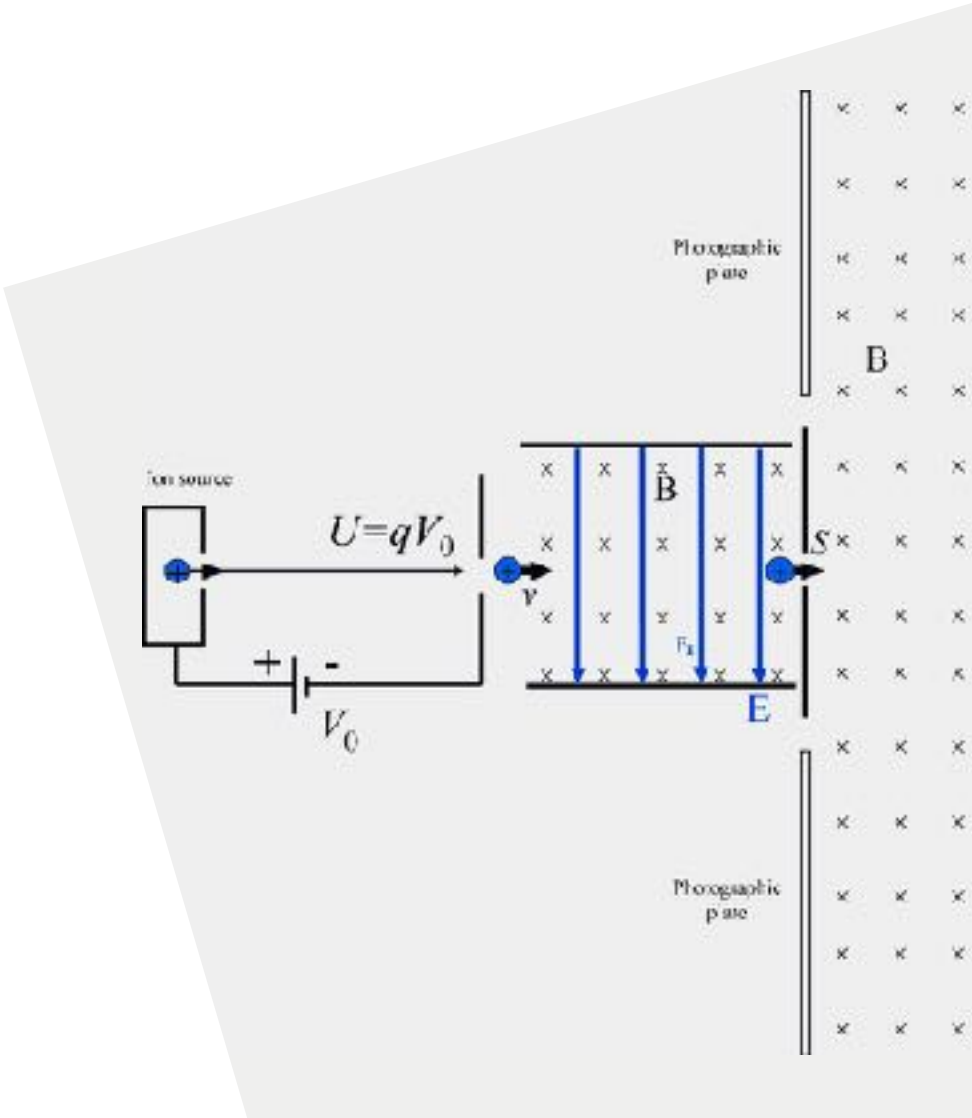
PHYC10009 Physics Foundations

Magnetism
Week 9 Lecture 1

Professor Phillip Urquijo
School of Physics
University of Melbourne



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- Can measure magnetic field direction and strength with a compass needle
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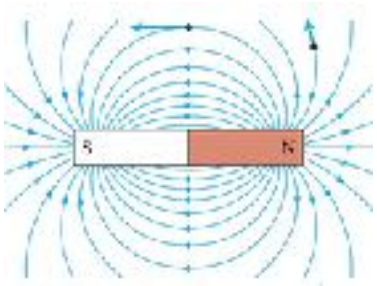

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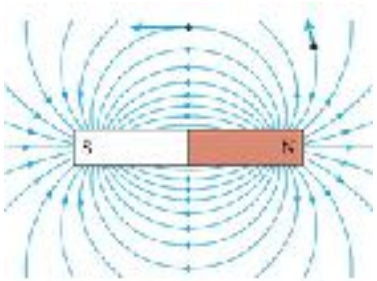

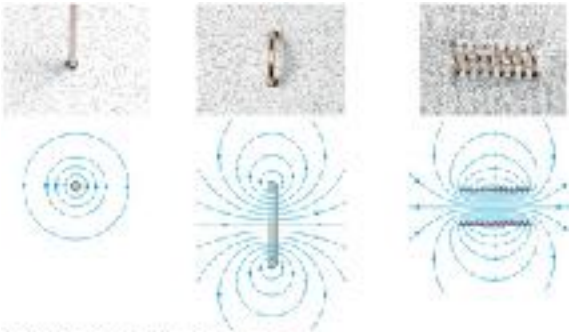

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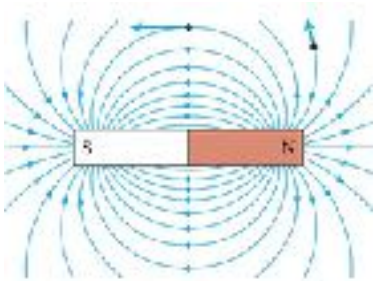

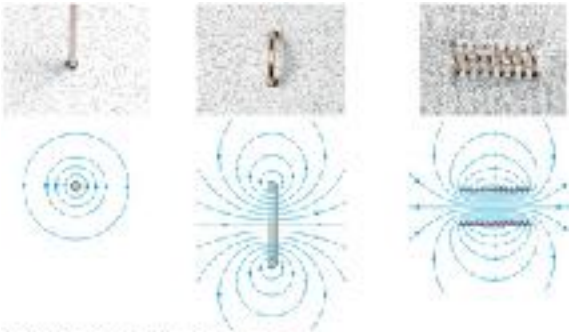

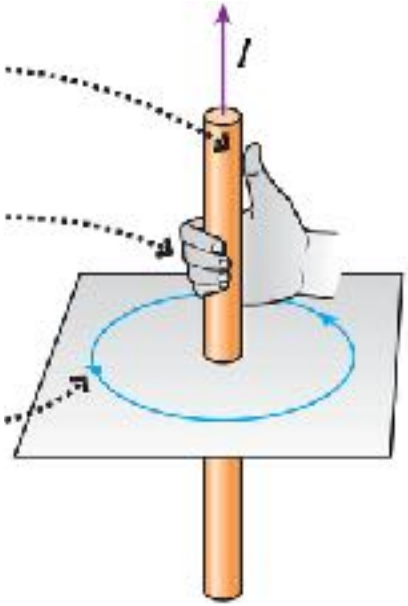

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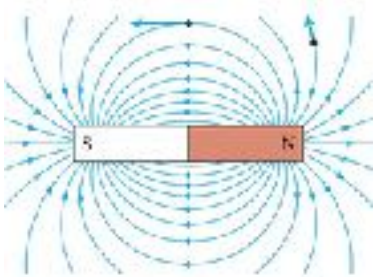

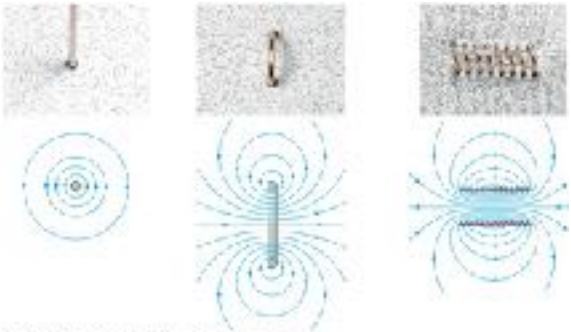

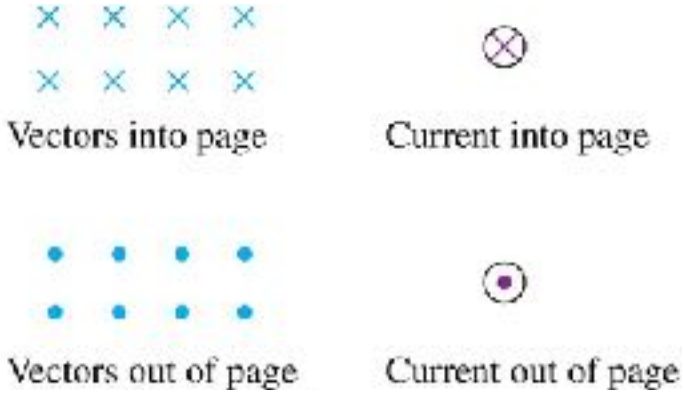

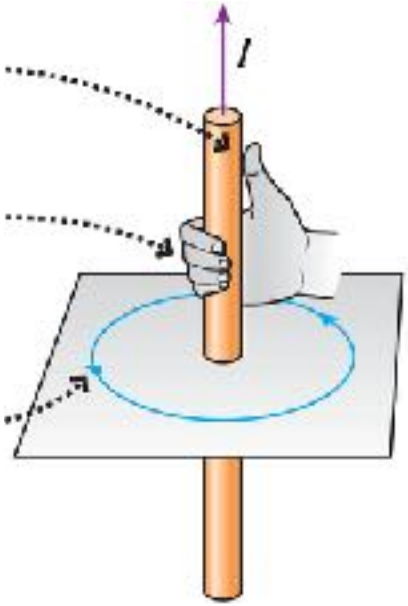

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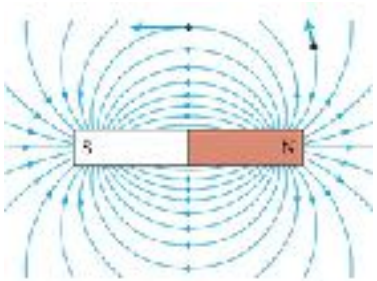
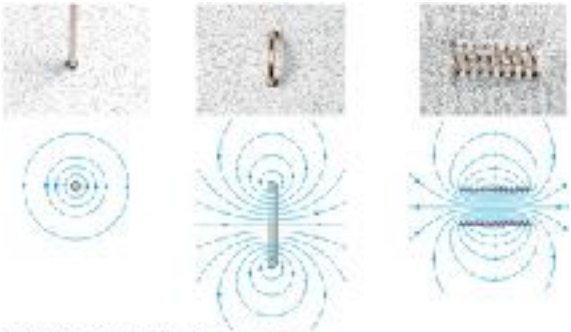
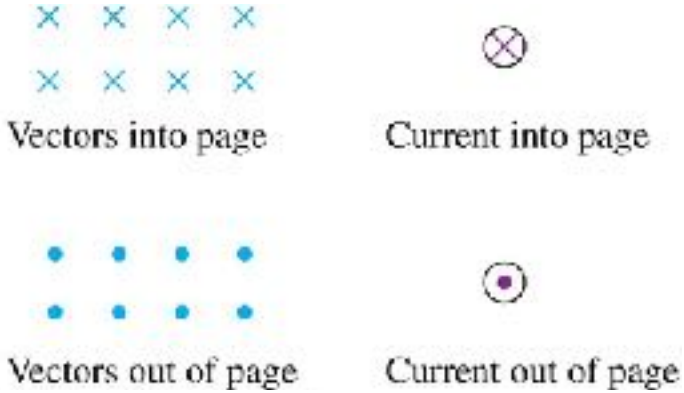
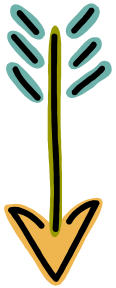
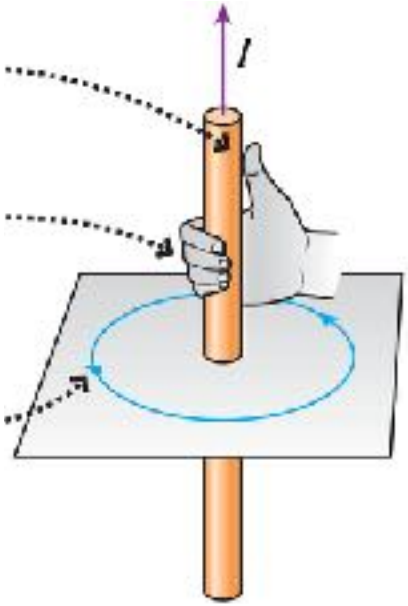










20.5 Quantifying and calculating magnetic fields

Magnetic field of a long straight wire:

$$B_{\text{wire}} = \frac{\mu_0 I}{2\pi r}$$

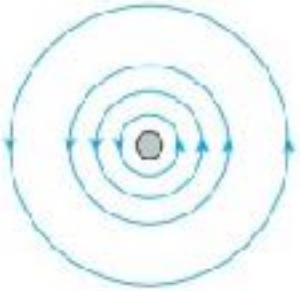
B = magnetic field in units of Tesla, T

μ_0= magnetic permeability

$4\pi \times 10^{-7}\text{Tm/A}$

I = electric current in amps

r = distance from the wire



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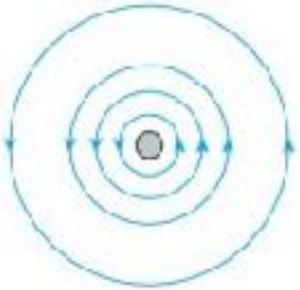

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Field Location	Field Strength (T)
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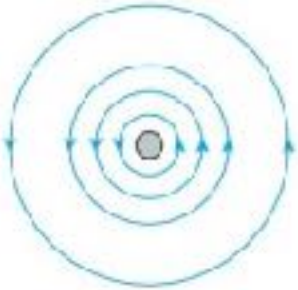

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Field Location	Field Strength (T)
Surface of the Earth	5×10^{-5} T
Fridge magnet	5×10^{-3} T
Lab magnet	$0.1 - 1.0$ T
MRI	$2 - 10$ T

Magnetic fields from more than one source

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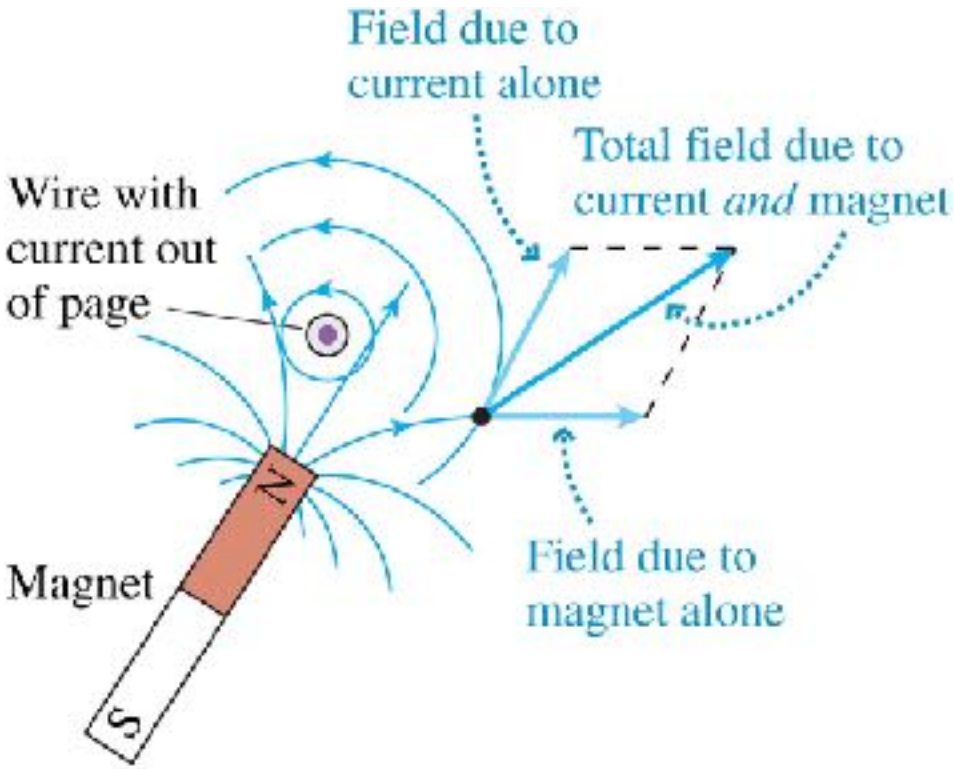

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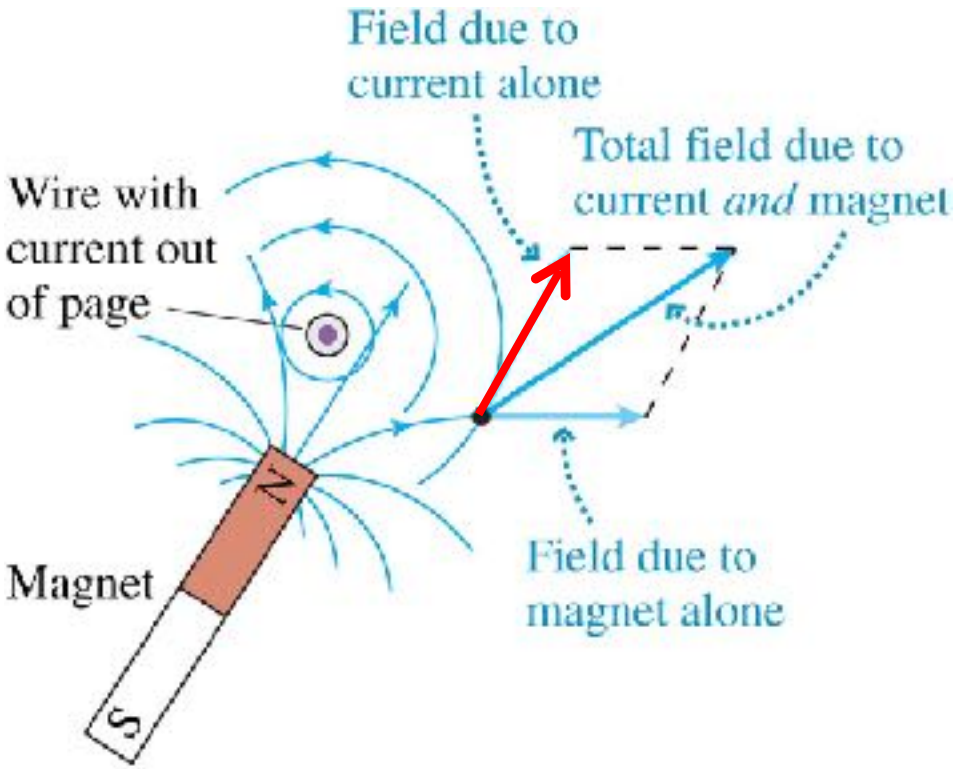

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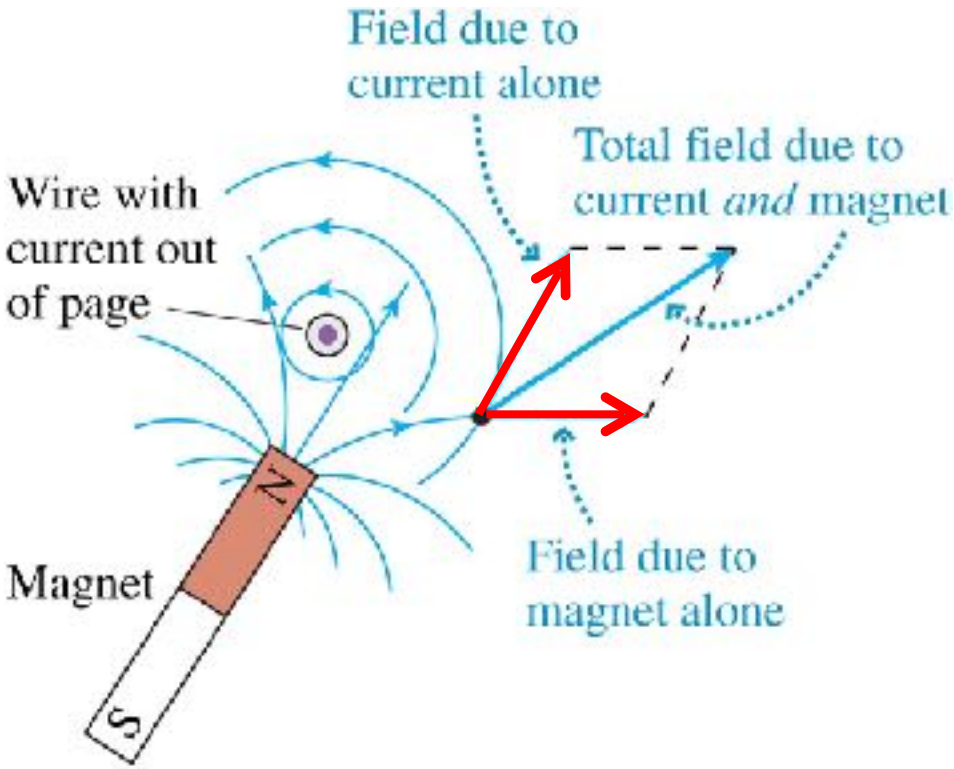

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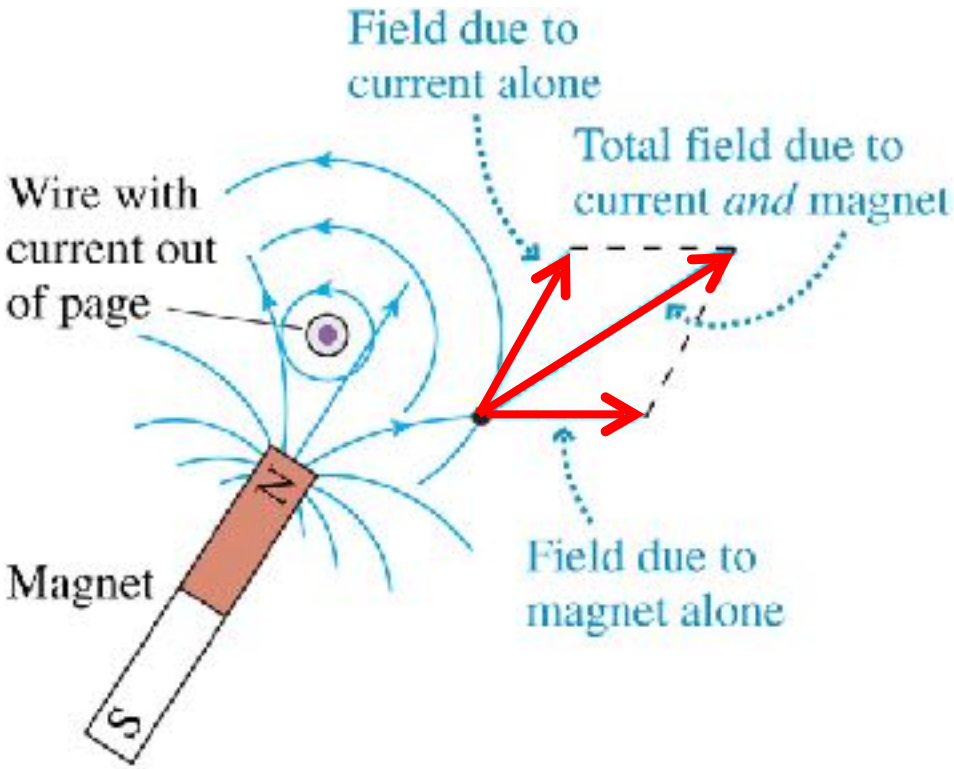

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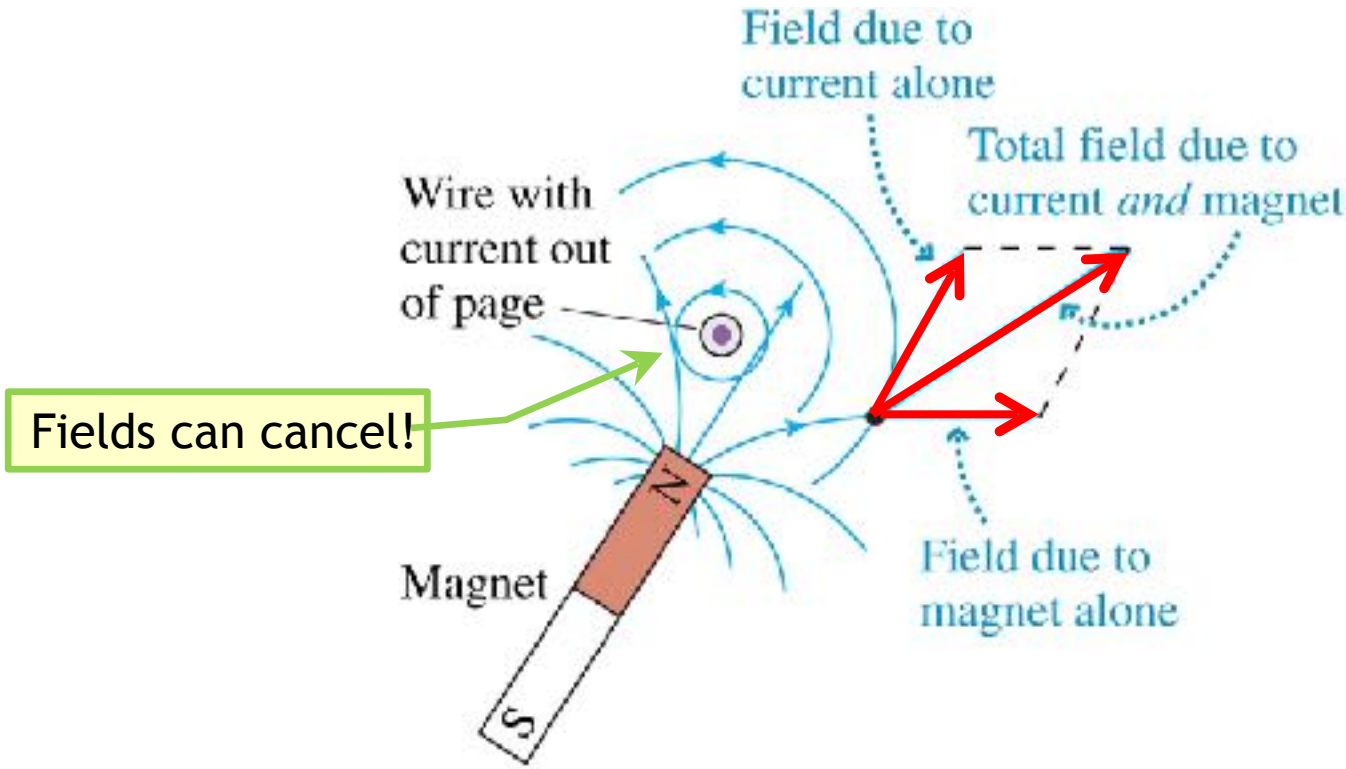
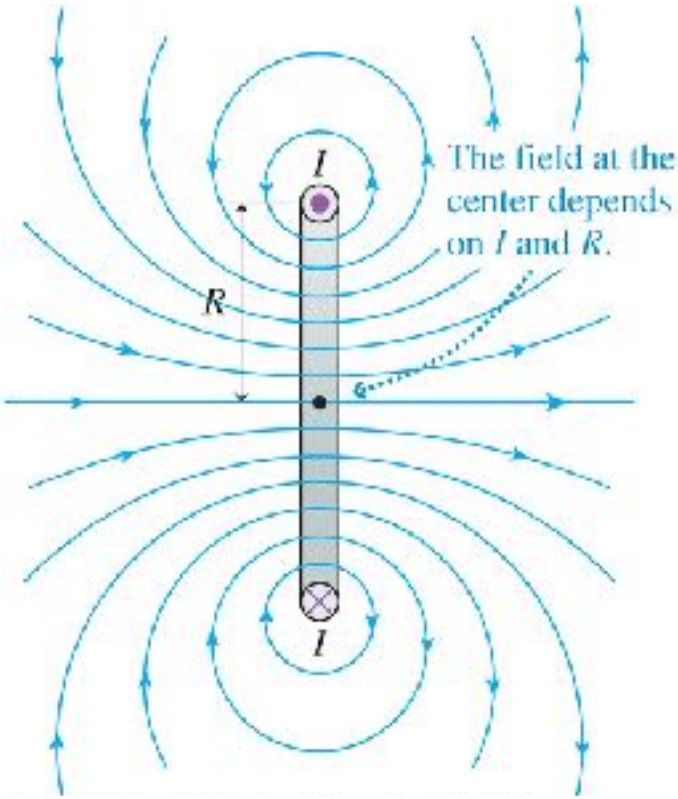


Magnetic field due to a current loop

- Wrap wire back on itself: make a loop
- Fields add up at centre
- Multiple loops (coil) = bigger field

- Note for later: a current-carrying loop is like a little bar-magnet



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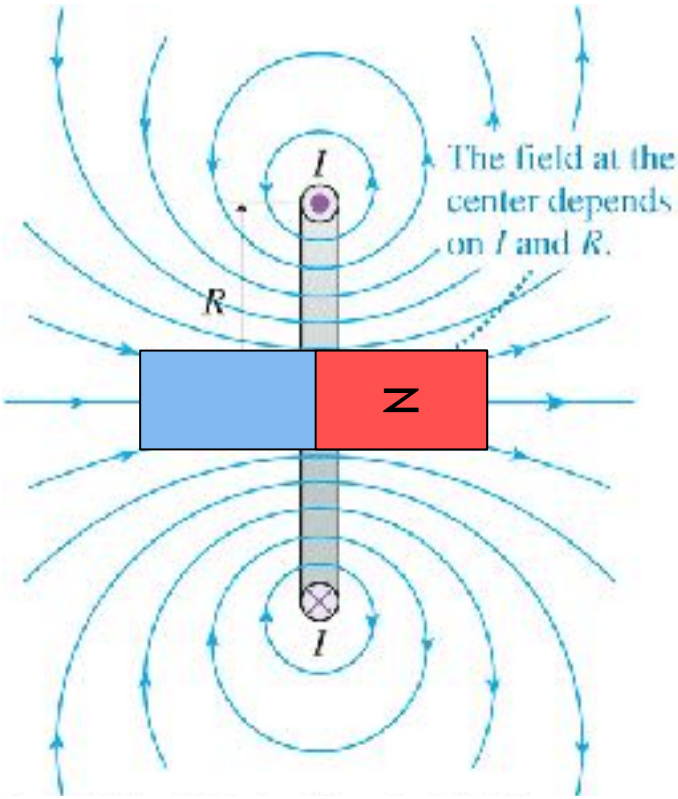
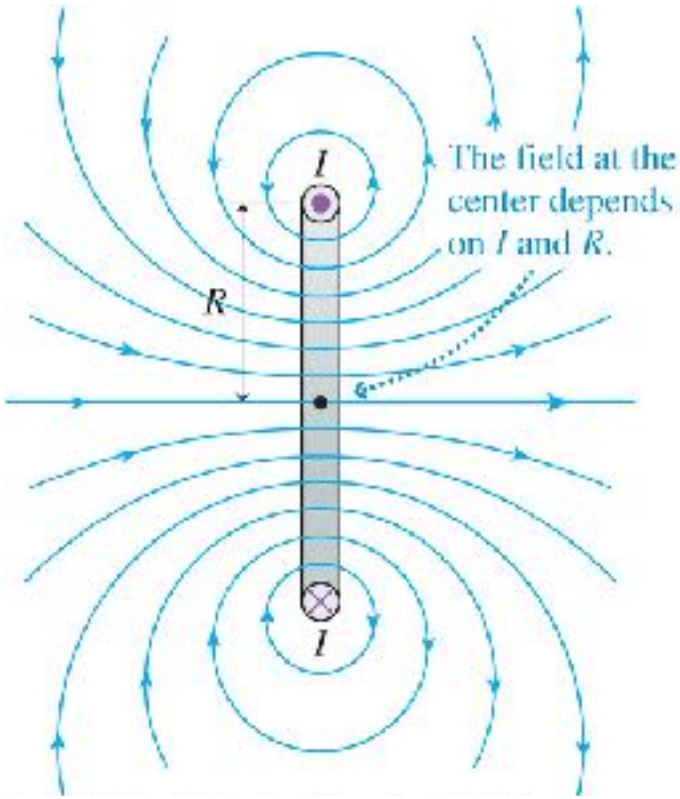


Magnetic field due to a current loop

- Magnetic field at centre of a current loop, radius r is $B = \dfrac{\mu_0 I}{2r}$
- Note no π in denominator
- If multiple coils or turns in the loop, each contributes the same amount:

$$B_{\text{loop}} = \frac{\mu_0 N I}{2r}$$

where N is the number of turns



20.7 Magnetic field due to a solenoid

- A solenoid is an extended current loop

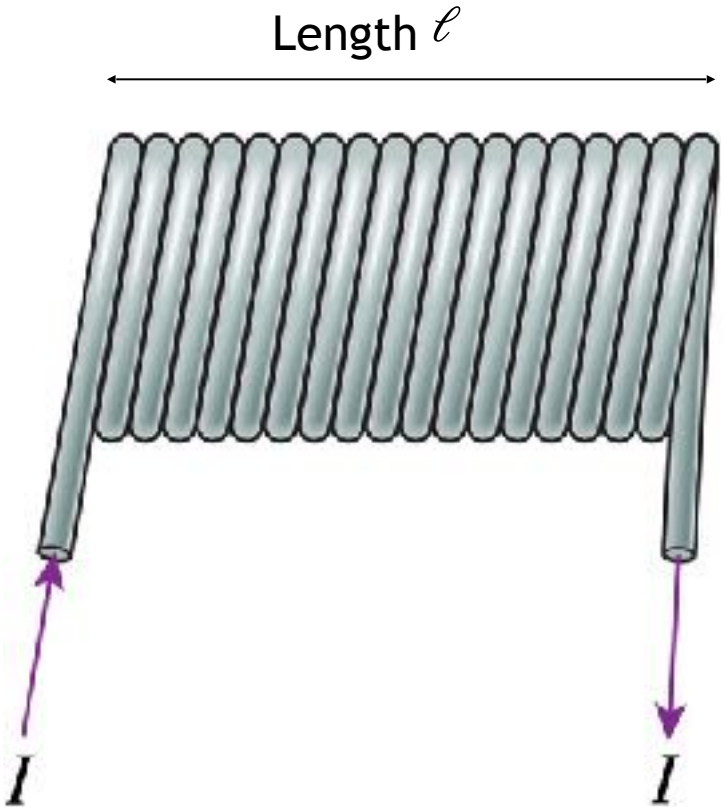
- Magnetic field <u>inside</u> the solenoid is

$$B_{\text{solenoid}} = \mu_0 I \frac{N}{\ell} = \mu_0 n I$$

where N = total number of turns

ℓ = length of solenoid

n = turns per unit length



- A solenoid is at the heart of a medical MRI machine

Diamagnetism (non-examinable)

Induced magnetic field opposite applied magnetic field. $B_D = -\mu_D B_{applied}$

Large enough field → force is greater than gravity

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D EH-10 Diamagnetism Levitation

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Dia(bolical)magnetism (non-examinable)

http://www.hfml.kun.nl/levitate.html; http://www.ru.nl/hfml/research/levitation/

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20 Tesla frog magnet

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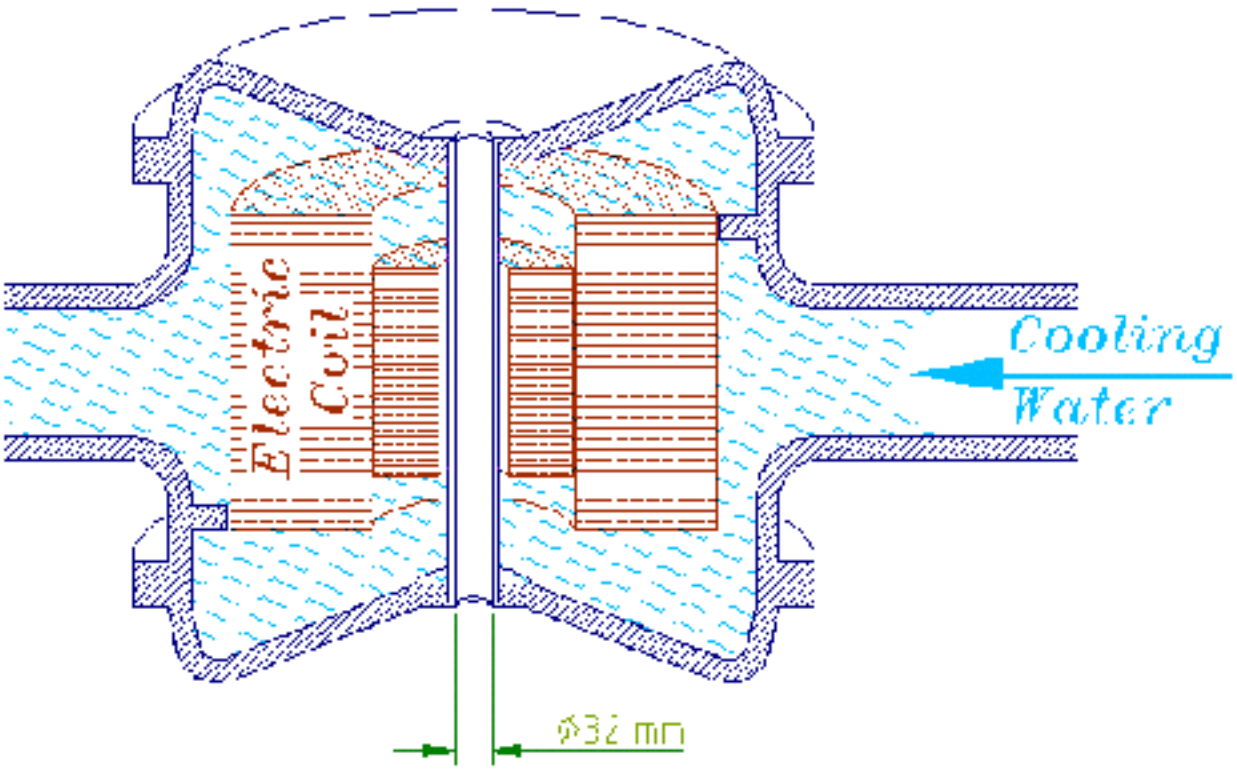
- 6000 kW (20kA, 300V)
  ($1500 per hour)
- Cooling water
  - 10°C
  - 300 m³/h
  - 12bar inlet pressure



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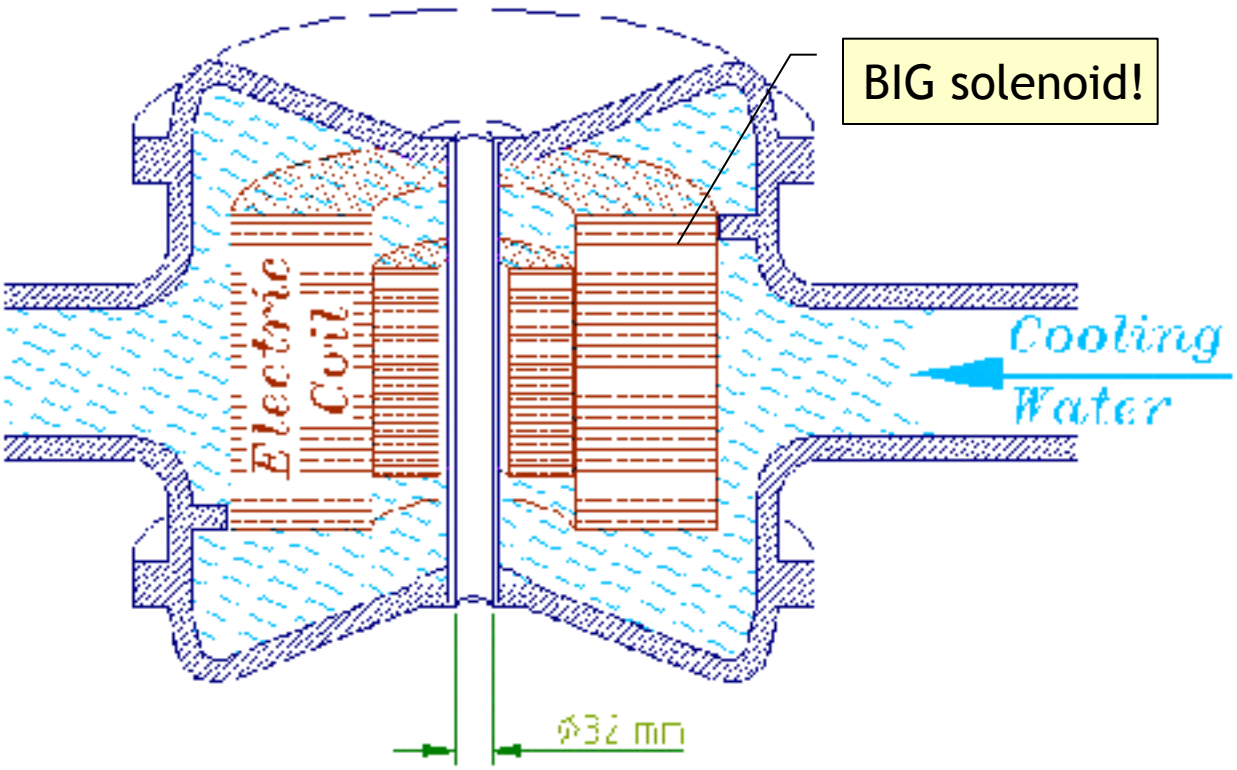
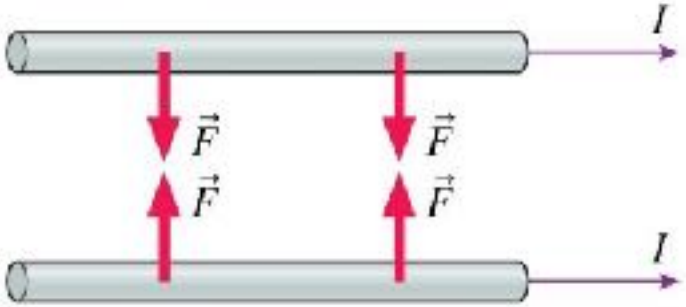


Magnetic force on *moving* charge

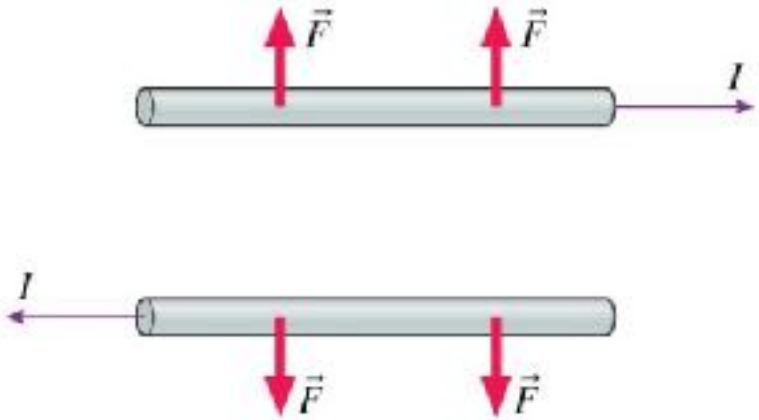
$$F = qvB$$

20.3 Magnetic forces on currents and charges

- Magnetic fields exert forces on moving charges, e.g. electric currents
- Since a current-carrying wire produces a magnetic field, there are magnetic forces between current-carrying wires



"Like" currents attract.



"Opposite" currents repel.

- We can understand that by first looking at the magnetic force on a moving charge

Demo Ea-1

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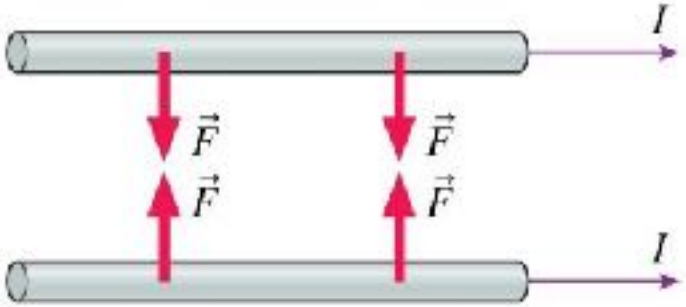

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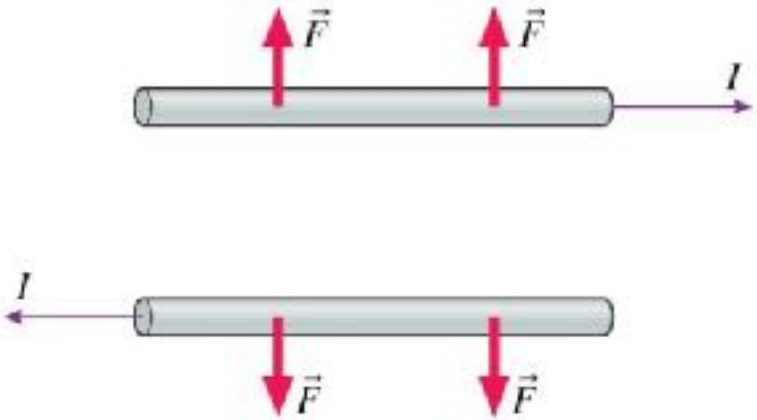

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D Ea-9 Attraction and Repulsion of Currents

Demo Ea-1

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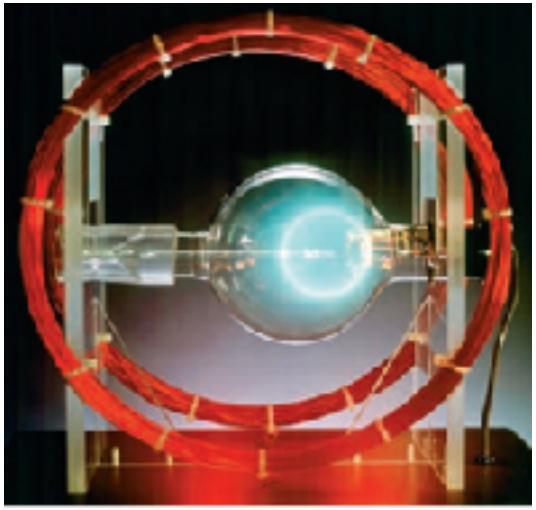
- Force on a particle of charge q moving with velocity $\vec{\mathbf{v}}$ in magnet field $\vec{\mathbf{B}}$:

$$F = qvB\sin\theta$$



- <u>*Magnitude*</u> of force is $F = qvB\sin\theta$ where θ is the angle between $\vec{v}$ and $\vec{B}$

- Direction of $\vec{\mathbf{F}}$ depends on *Right Hand Rule*
  - $\vec{\mathbf{F}}$ is ***not*** in the direction of $\vec{\mathbf{B}}$!!
  - If velocity is <u>*perpendicular*</u> to $\vec{B}$ then $\theta = 90°$ and $F = qvB$

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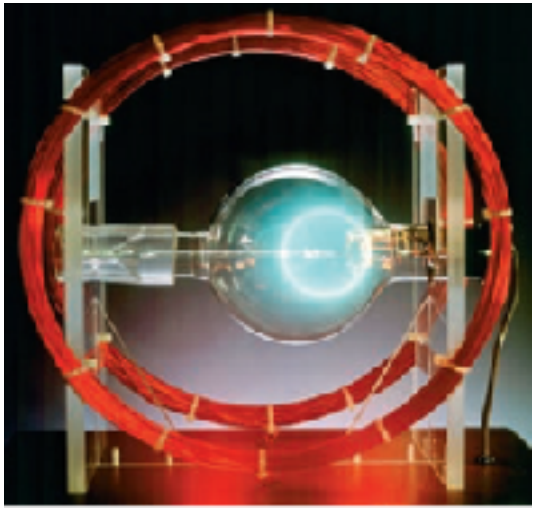

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D Ef-2 Bending of an electron beam

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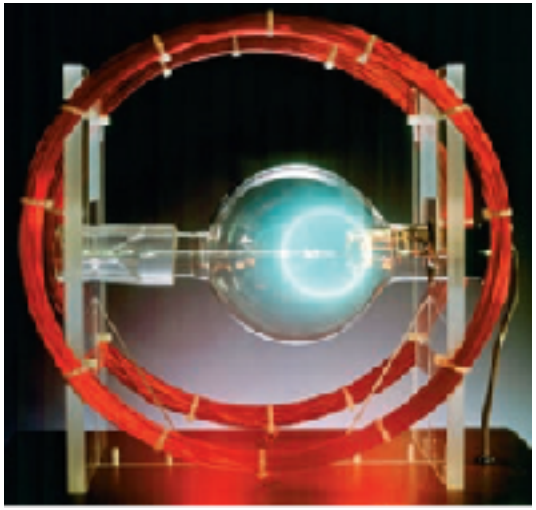

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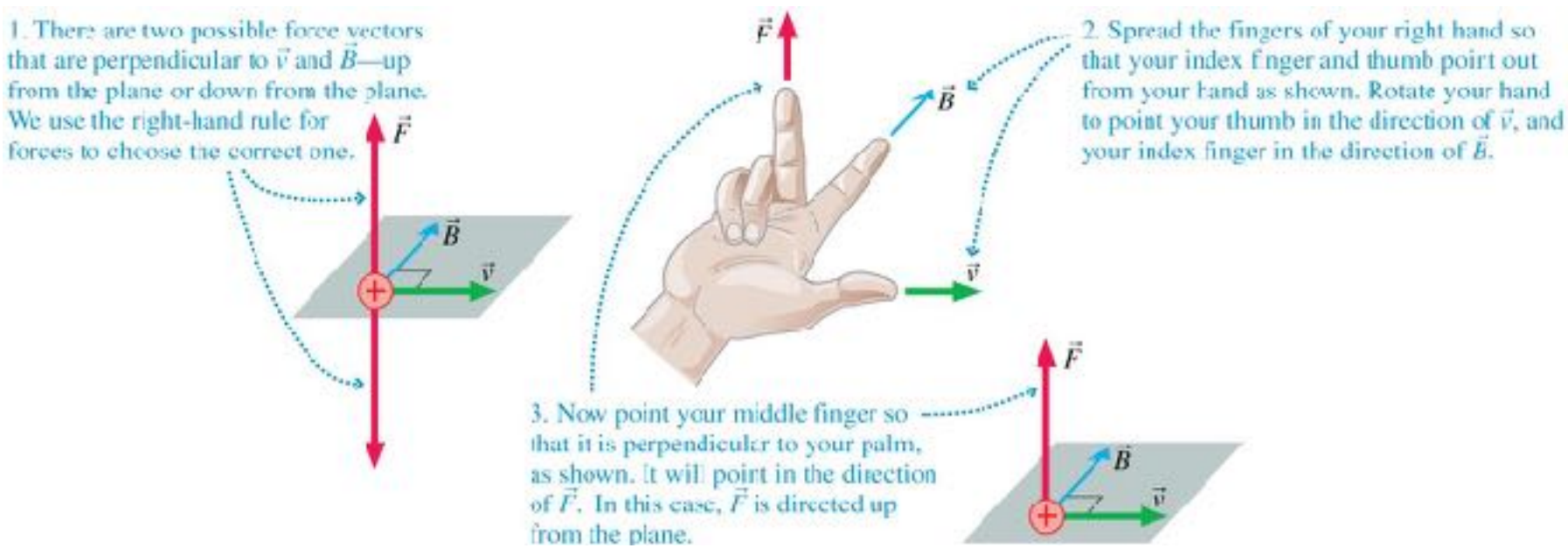
D Ea-1 Force on a Current Carrying Conductor in a magnetic field

D Ef-2 Bending of an electron beam

Right hand rule

- Force on a particle of charge q moving with velocity $\vec{\mathbf{v}}$ in magnet field $\vec{\mathbf{B}}$:

$$F = qvB\sin\theta \qquad \vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$



$\vec{F}$ is Perpendicular to both $\vec{v}$ and $\vec{B}$ as given by the Right Hand Rule

Right Hand Rule

- Force on a particle of charge q moving with velocity $\vec{\mathbf{v}}$ in magnet field $\vec{\mathbf{B}}$:

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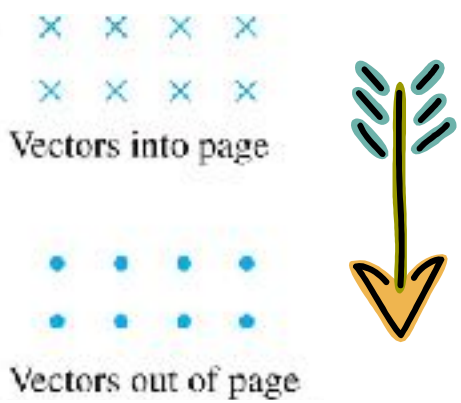
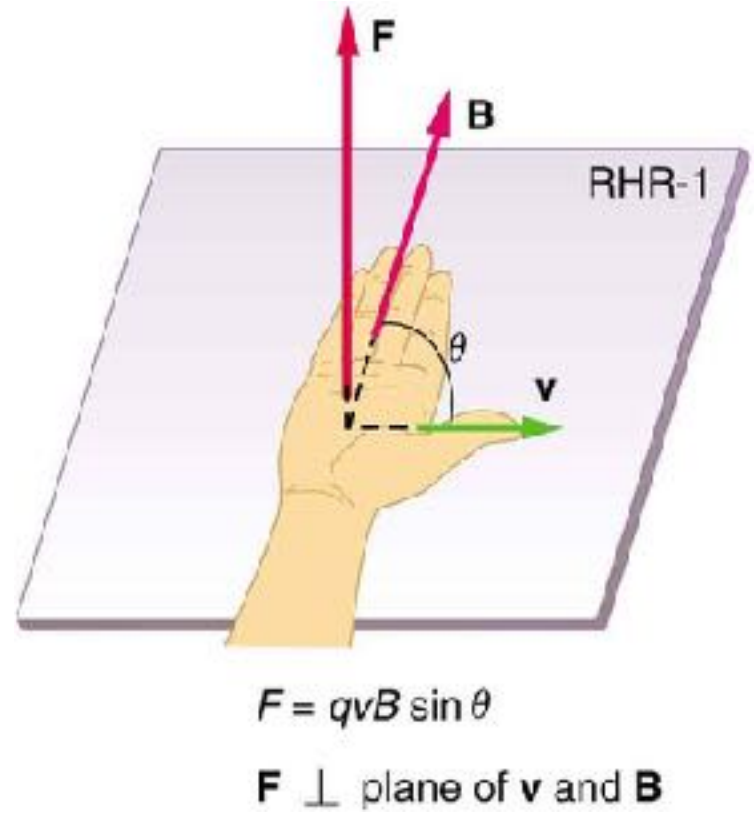




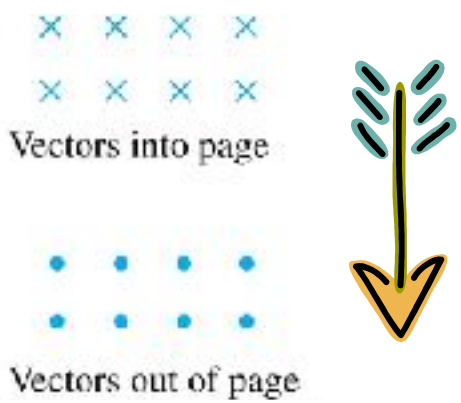
$\vec{F}$ is Perpendicular to both $\vec{v}$ and $\vec{B}$ as given by the Right Hand Rule

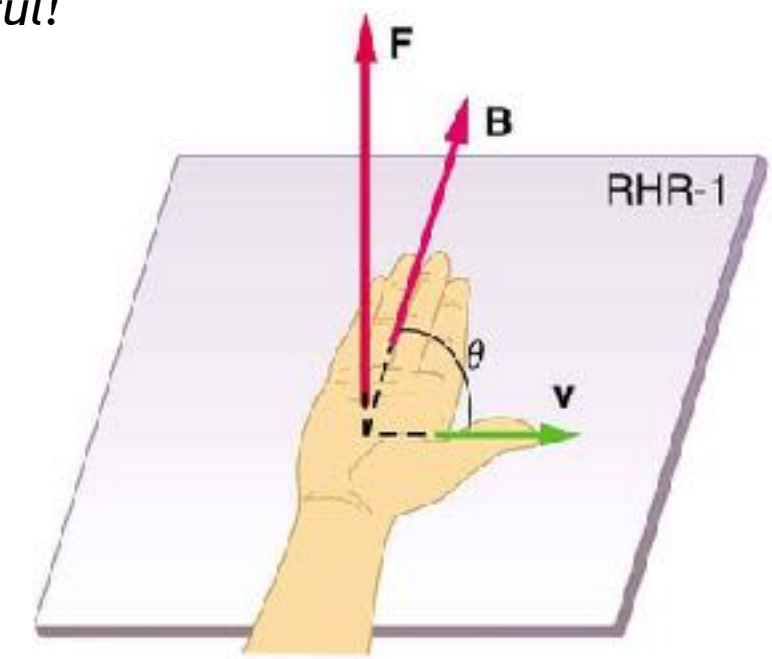
Right Hand Rule

- Force on a particle of charge q moving with velocity $\vec{\mathbf{v}}$ in magnet field $\vec{\mathbf{B}}$:

$$F = qvB\sin\theta \qquad \vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

Very Bloody Forceful!





$F = qvB\sin\theta$

F ⊥ plane of **v** and **B**

$\vec{F}$ is Perpendicular to both $\vec{v}$ and $\vec{B}$ as given by the Right Hand Rule

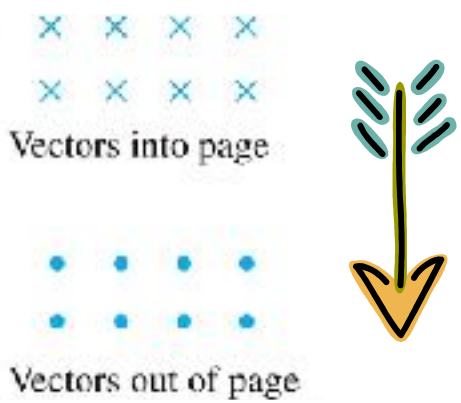
Right Hand Rule

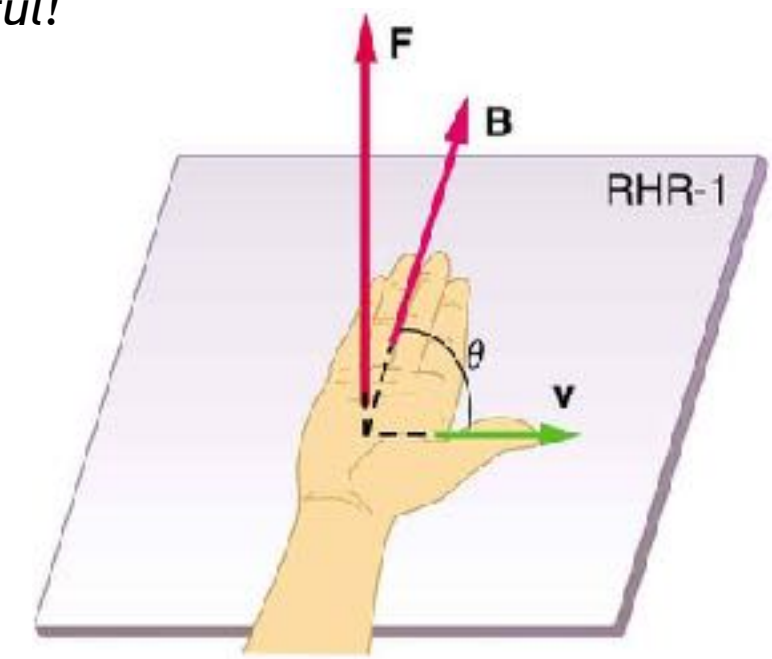
- Force on a particle of charge q moving with velocity $\vec{\mathbf{v}}$ in magnet field $\vec{\mathbf{B}}$:

$$F = qvB\sin\theta \qquad \vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

Very Bloody Forceful!

1. Thumb in direction of charged particle





$F = qvB\sin\theta$

F ⊥ plane of **v** and **B**

$\vec{F}$ is Perpendicular to both $\vec{v}$ and $\vec{B}$ as given by the Right Hand Rule

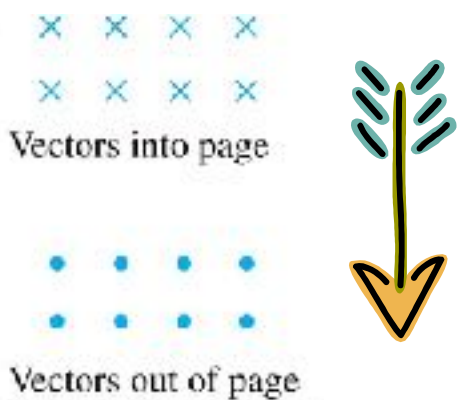
Right Hand Rule

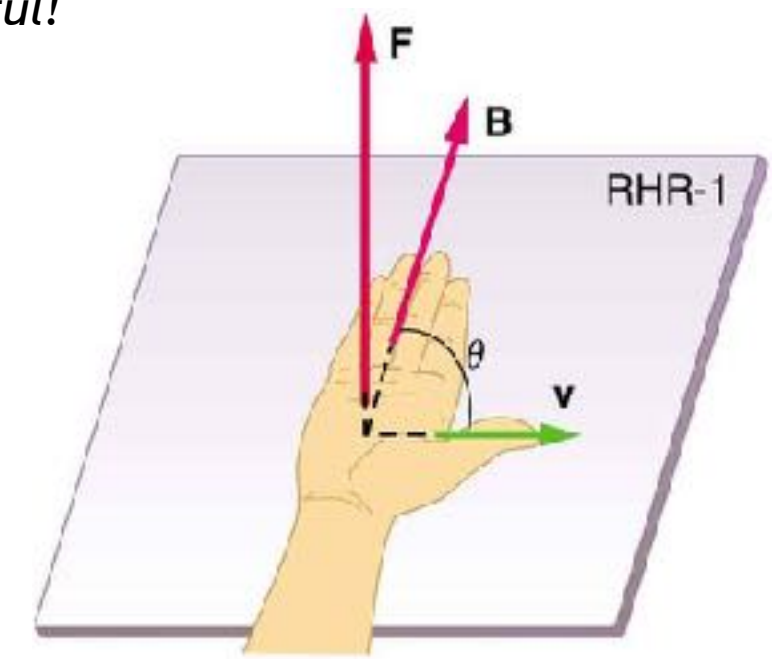
- Force on a particle of charge q moving with velocity $\vec{\mathbf{v}}$ in magnet field $\vec{\mathbf{B}}$:

$$F = qvB\sin\theta \qquad \vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

Very Bloody Forceful!

1. Thumb in direction of charged particle
2. Fingers in direction of Magnetic field





$F = qvB\sin\theta$

F ⊥ plane of **v** and **B**

$\vec{F}$ is Perpendicular to both $\vec{v}$ and $\vec{B}$ as given by the Right Hand Rule

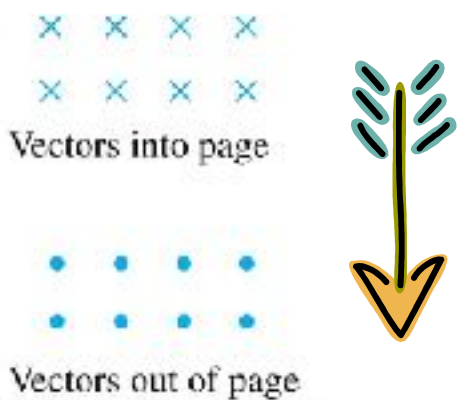
Right Hand Rule

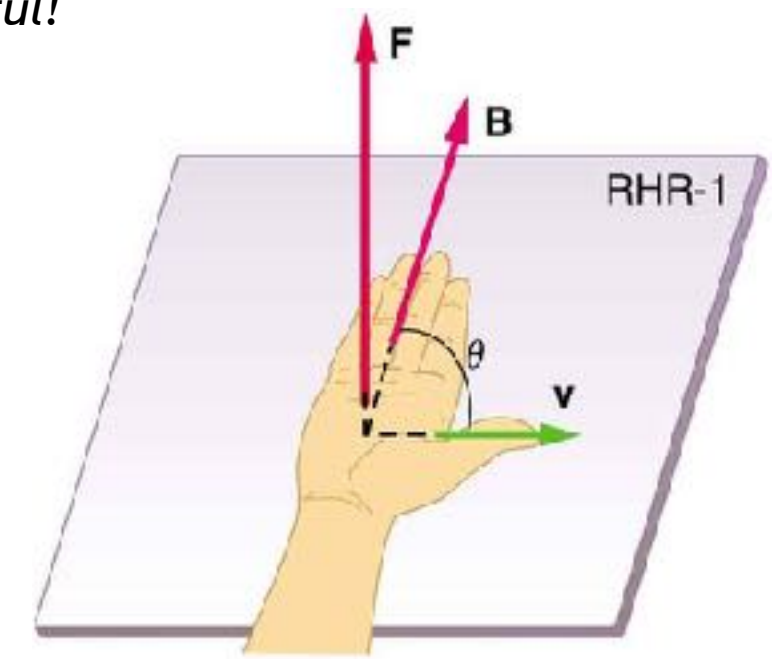
- Force on a particle of charge q moving with velocity $\vec{\mathbf{v}}$ in magnet field $\vec{\mathbf{B}}$:

$$F = qvB\sin\theta \qquad \vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

Very Bloody Forceful!

1. Thumb in direction of charged particle
2. Fingers in direction of Magnetic field
3. Force in the direction of a palm slap!





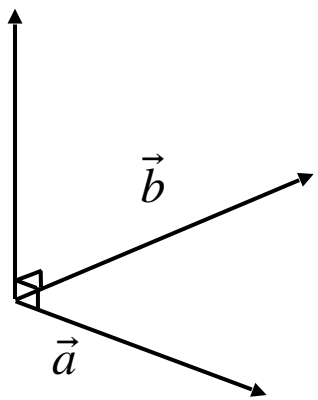
$F = qvB\sin\theta$

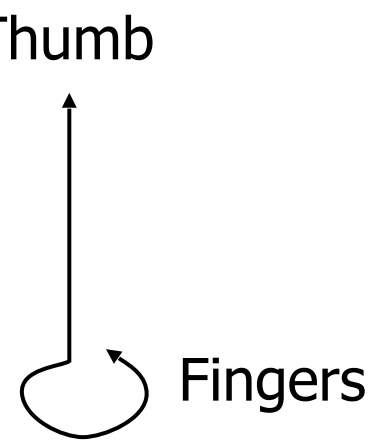
F ⊥ plane of **v** and **B**

$\vec{F}$ is Perpendicular to both $\vec{v}$ and $\vec{B}$ as given by the Right Hand Rule

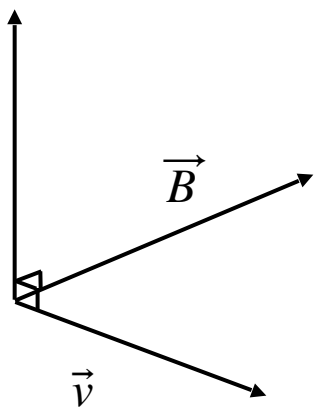
Vector products

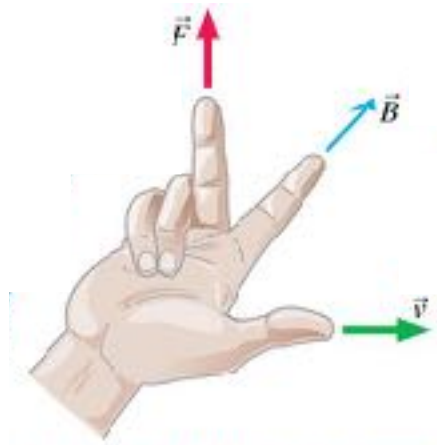
$\vec{a} \times \vec{b}$

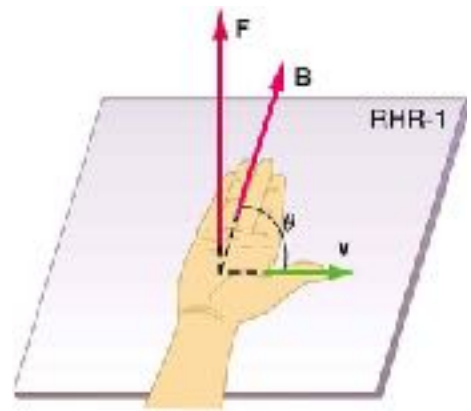




$F = q\vec{v} \times \vec{B}$



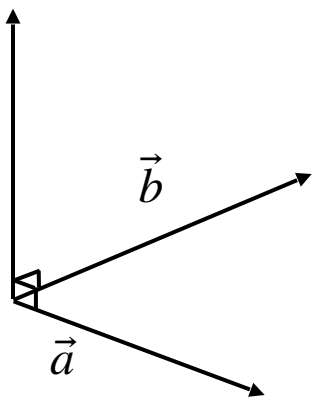


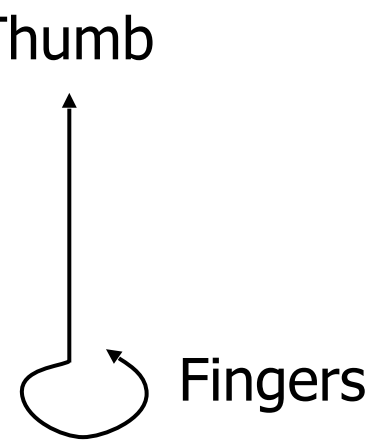


Vector products

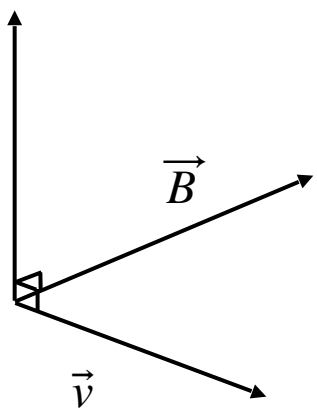
Note: direction for *positive* q
Opposite for negative q

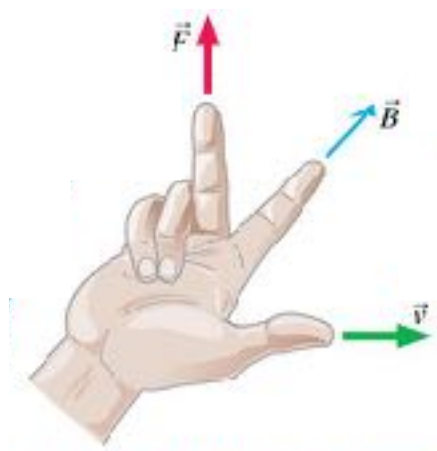
$\vec{a} \times \vec{b}$

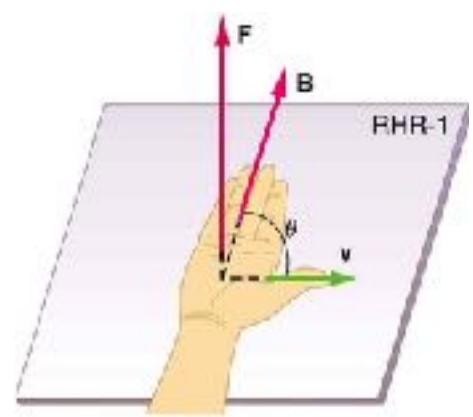




$F = q\vec{v} \times \vec{B}$





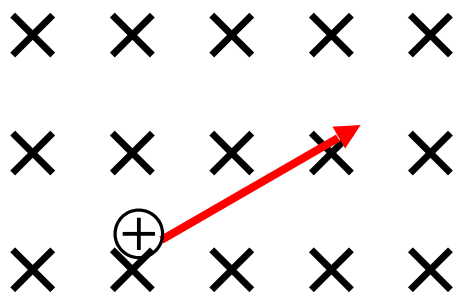


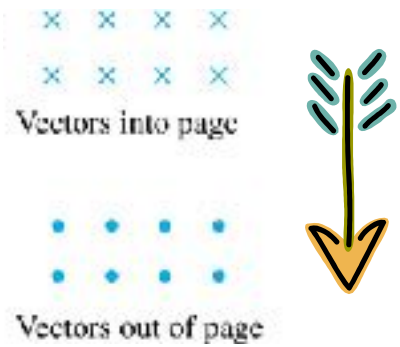
Direction of force

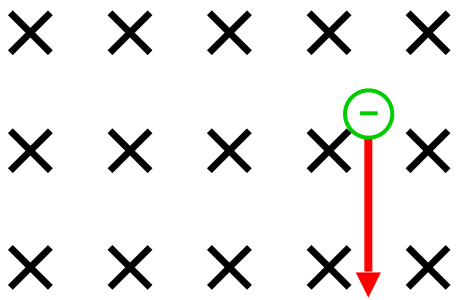
$$F = qvB\sin\theta$$

$$\vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

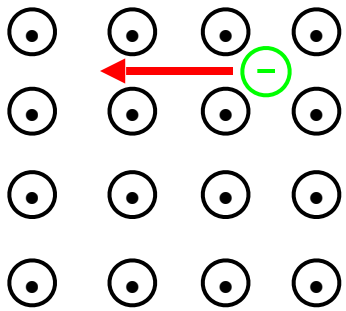
Example 1



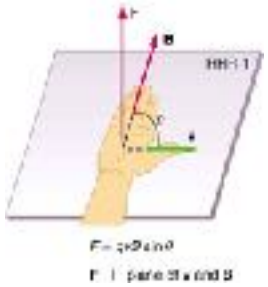


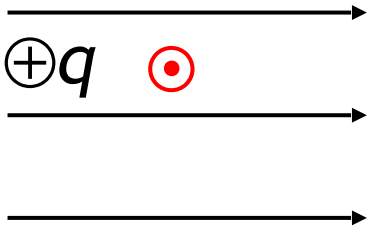


Example 2



Example 3





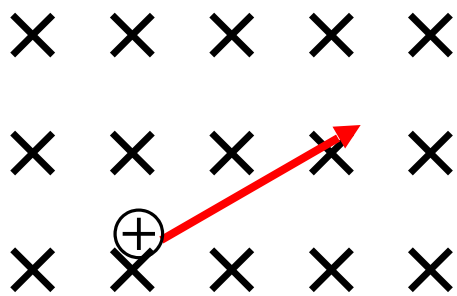
Example 4

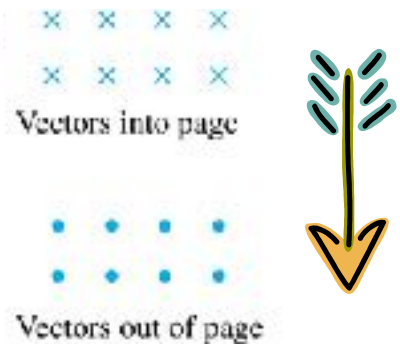
Direction of force

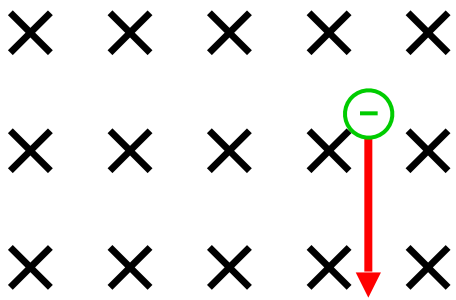
$$F = qvB\sin\theta$$

$$\vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

Example 1

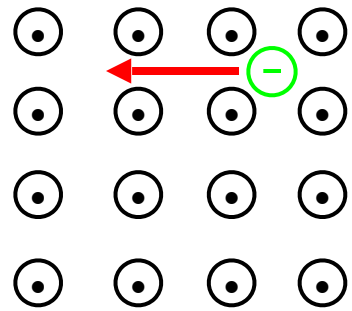




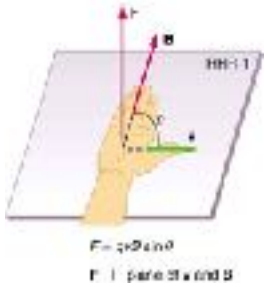


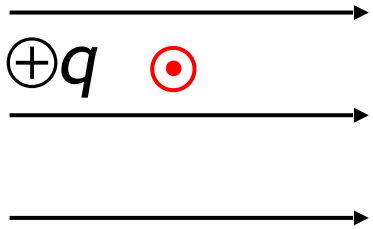
Example 2

Force is always perpendicular to both $\vec{\mathbf{v}}$ and $\vec{\mathbf{B}}$



Example 3





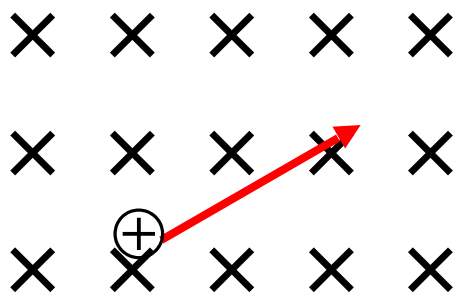
Example 4

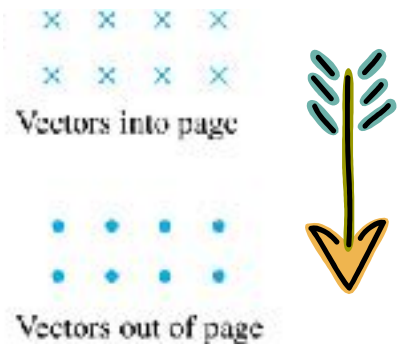
Vector products

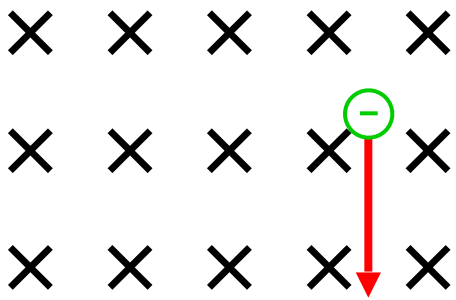
$$F = qvB\sin\theta$$

$$\vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

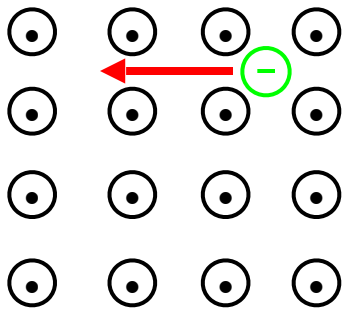
Example 1



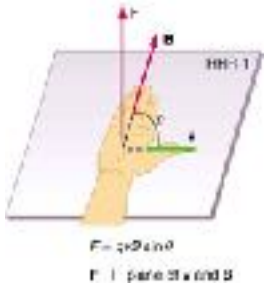


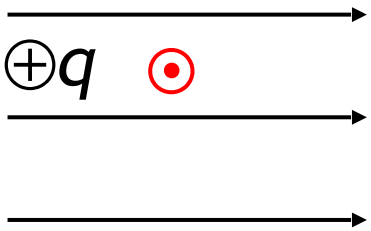


Example 2



Example 3





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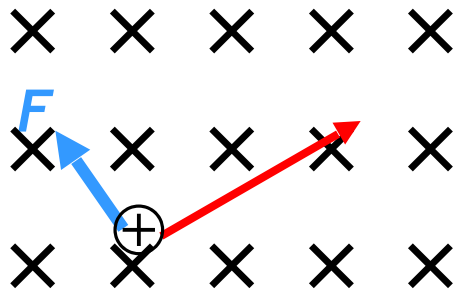
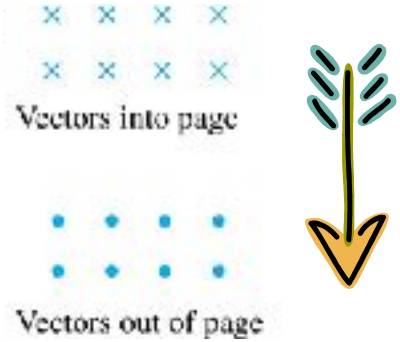
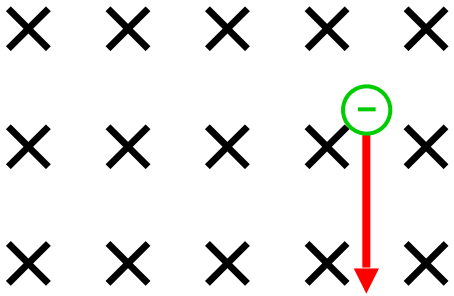
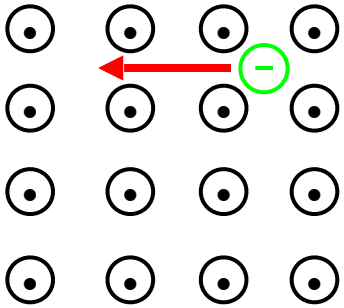


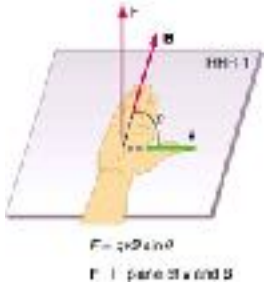


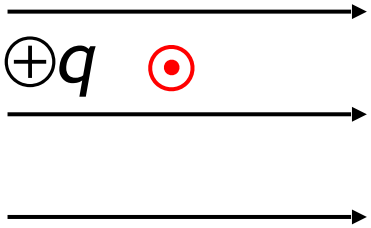


Example 2



Example 3





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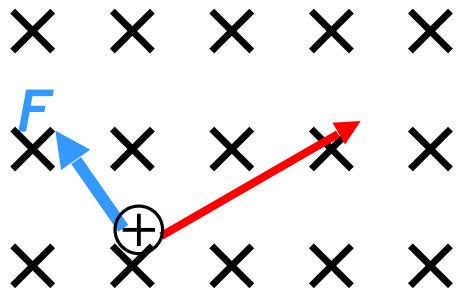
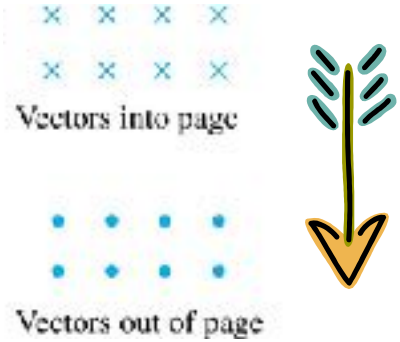
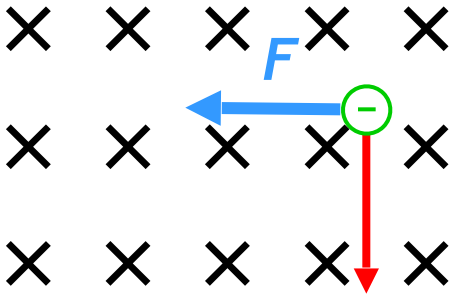
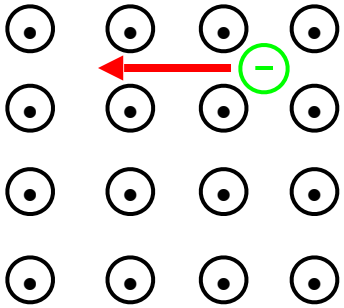


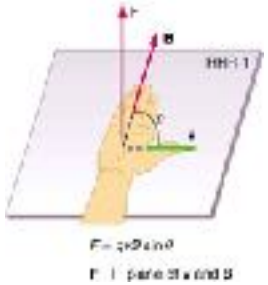


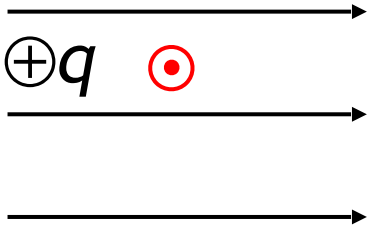


Example 2



Example 3



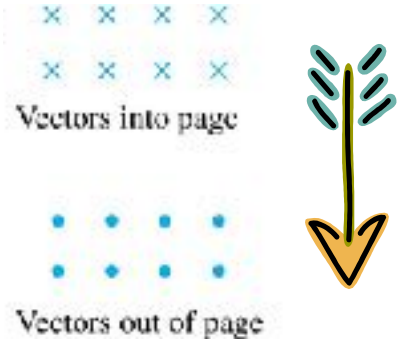


Example 4

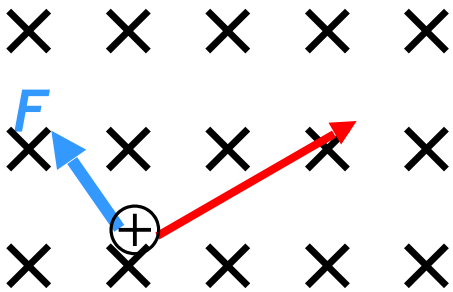
Vector products

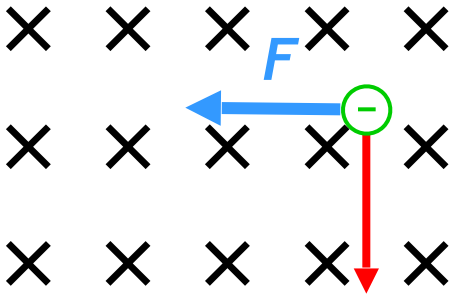
$$F = qvB\sin\theta$$

$$\vec{\mathbf{F}} = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

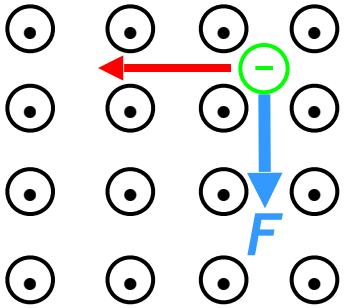


Example 1





Example 2



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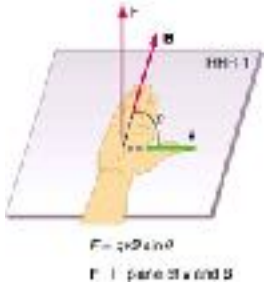

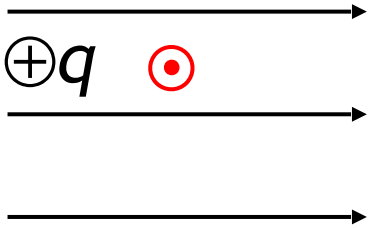

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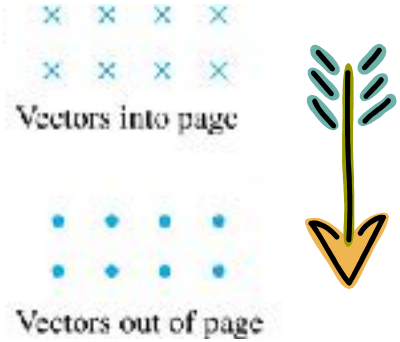

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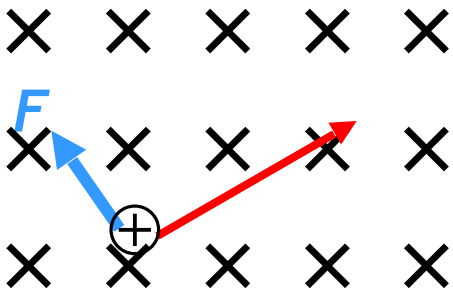
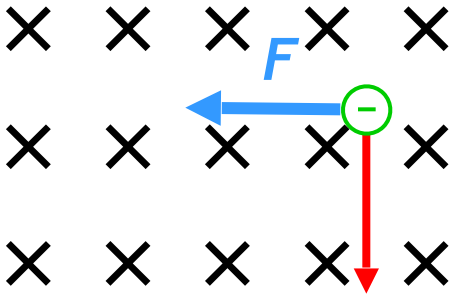
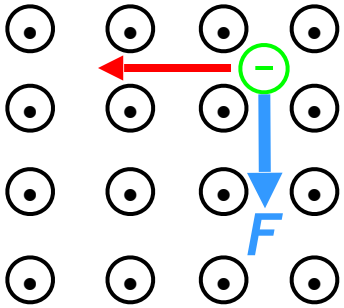




Example 2



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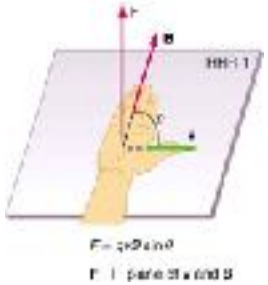

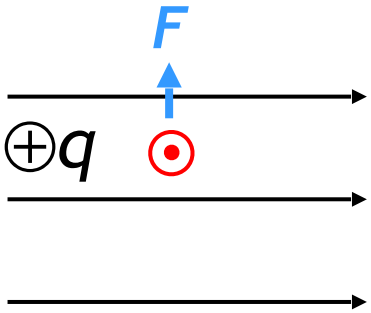

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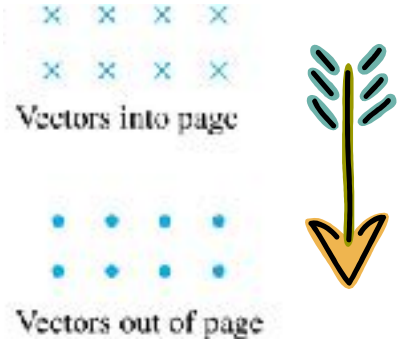

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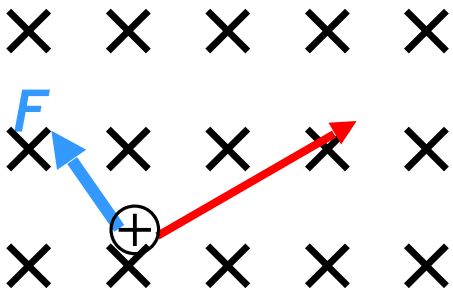
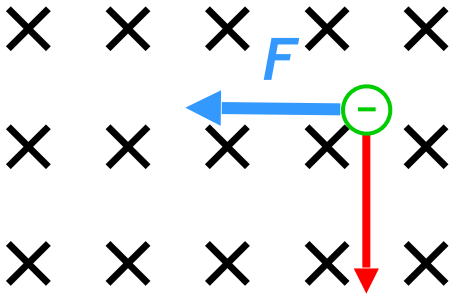
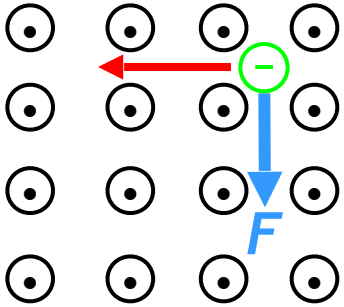


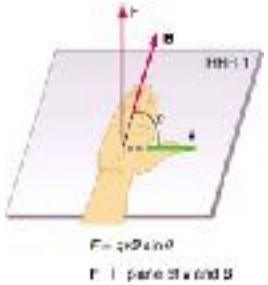


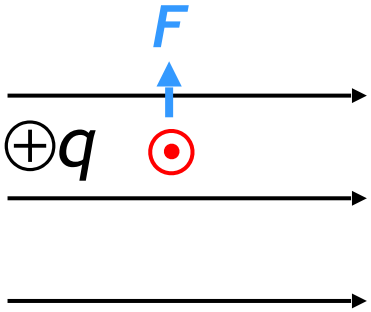
Example 2

Force is always perpendicular to both $\vec{\mathbf{v}}$ and $\vec{\mathbf{B}}$



Example 3



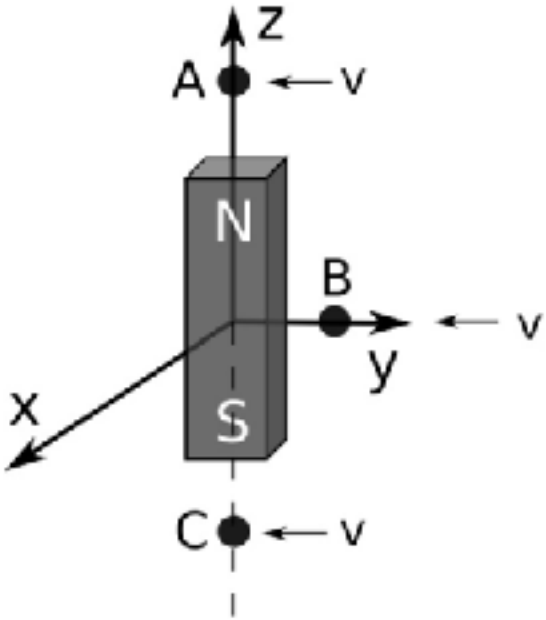


Example 4

Example: 2012 exam

Question 7 **[3 + 4 + 4 + 3 = 14 marks]**

Consider electrons travelling at velocity v from right to left near a bar magnet in a vacuum, with the bar magnet oriented along the z axis, and centred on the origin (see figure).

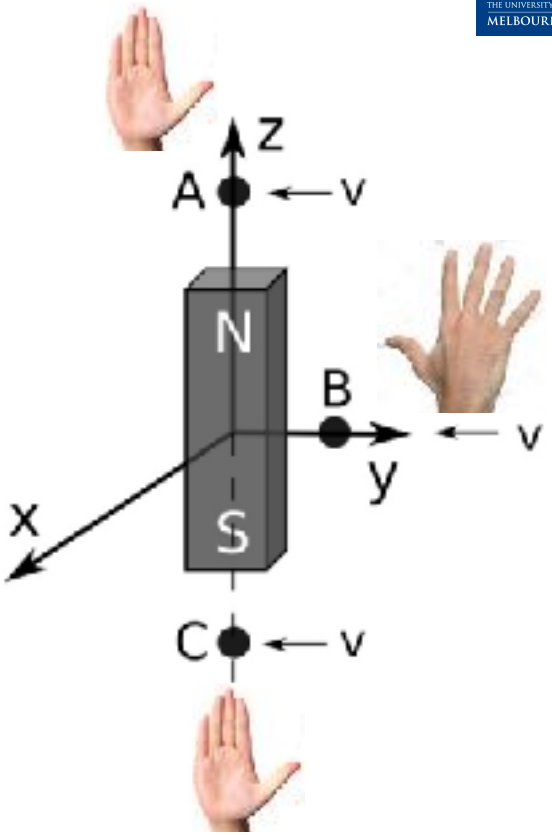


(a) Specify the direction of the force on the electrons at the points A, B and C which are all located in the yz plane (x=0). Assume that the electron paths are all parallel to the y axis at the three points, as shown by the vectors labelled v.

(b) At which point will the force have greater magnitude, A or B? Explain.

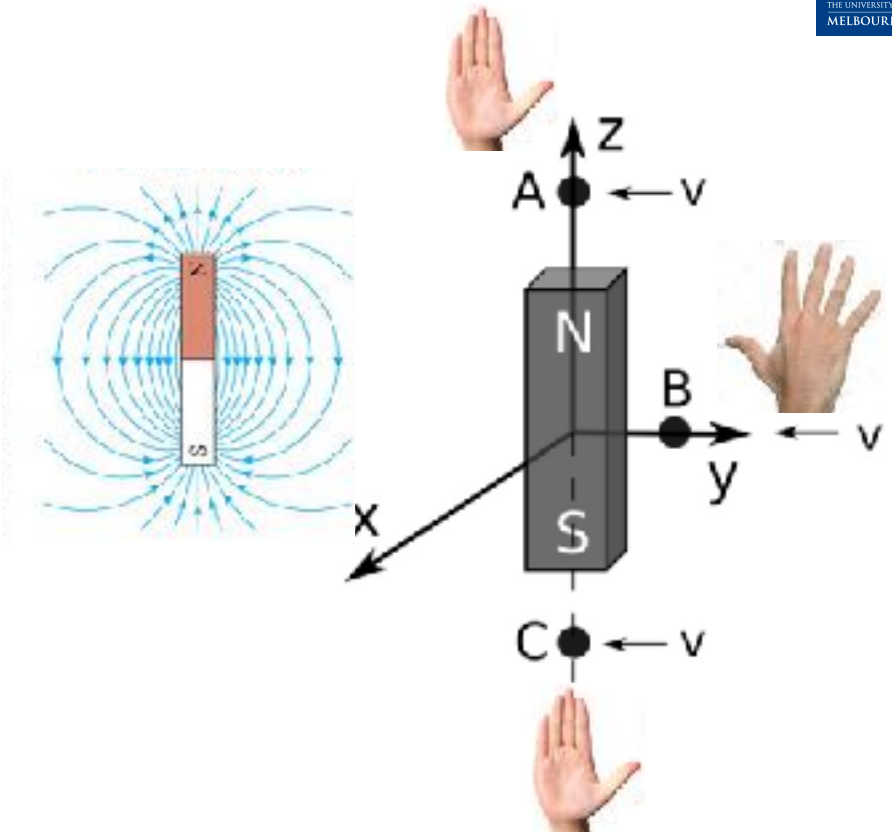
(d) The bar magnet is replaced with a loop of current-carrying wire which produces the same magnetic field. Sketch how the loop would be oriented, and indicate the direction of the current in the loop.

2012 Exam Solutions



(d)

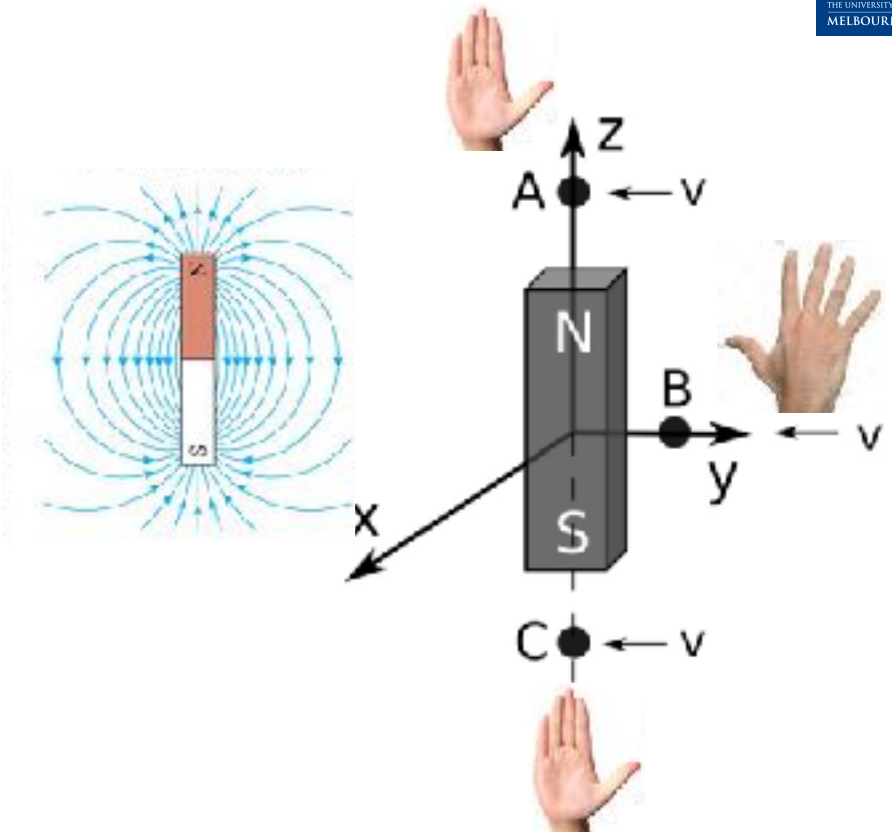
2012 Exam Solutions



(d)

2012 Exam Solutions

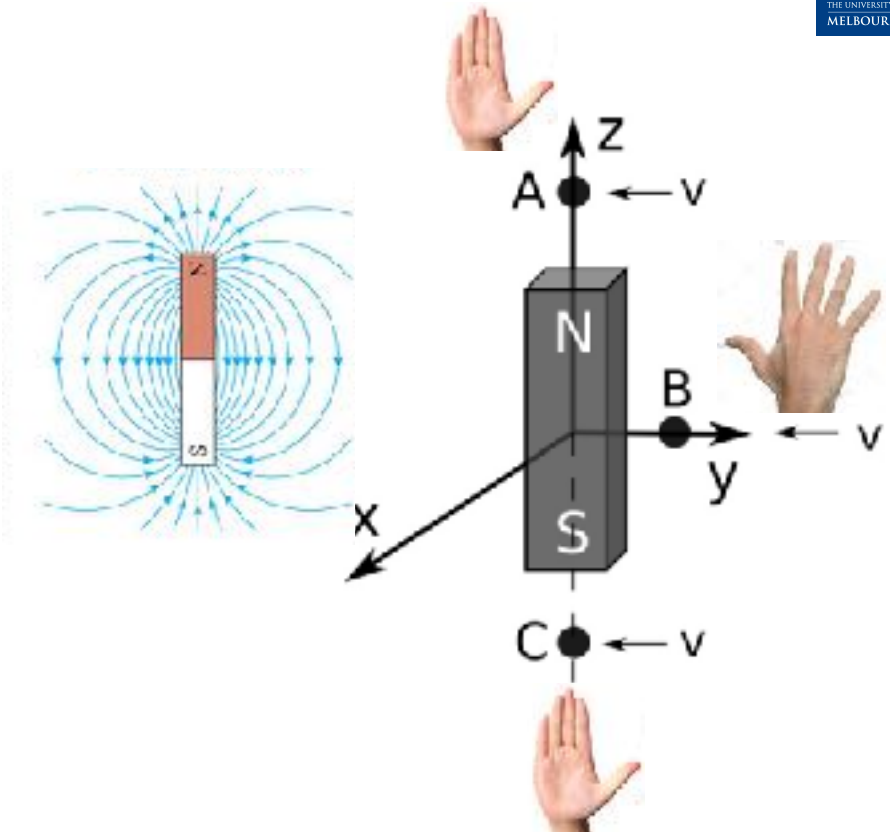
- (a) **A** $+\hat{x}$



(d)

2012 Exam Solutions

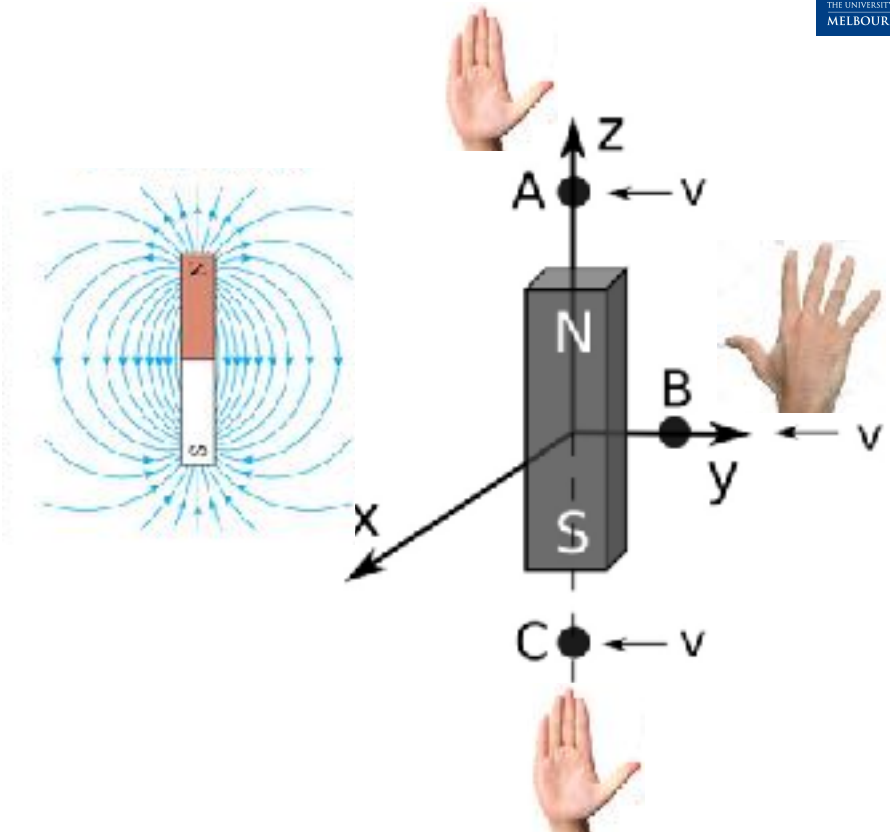
- (a) **A** $+\hat{x}$ **B** $-\hat{x}$



(d)

2012 Exam Solutions

- (a) **A** $+\hat{x}$ **B** $-\hat{x}$ **C** $+\hat{x}$



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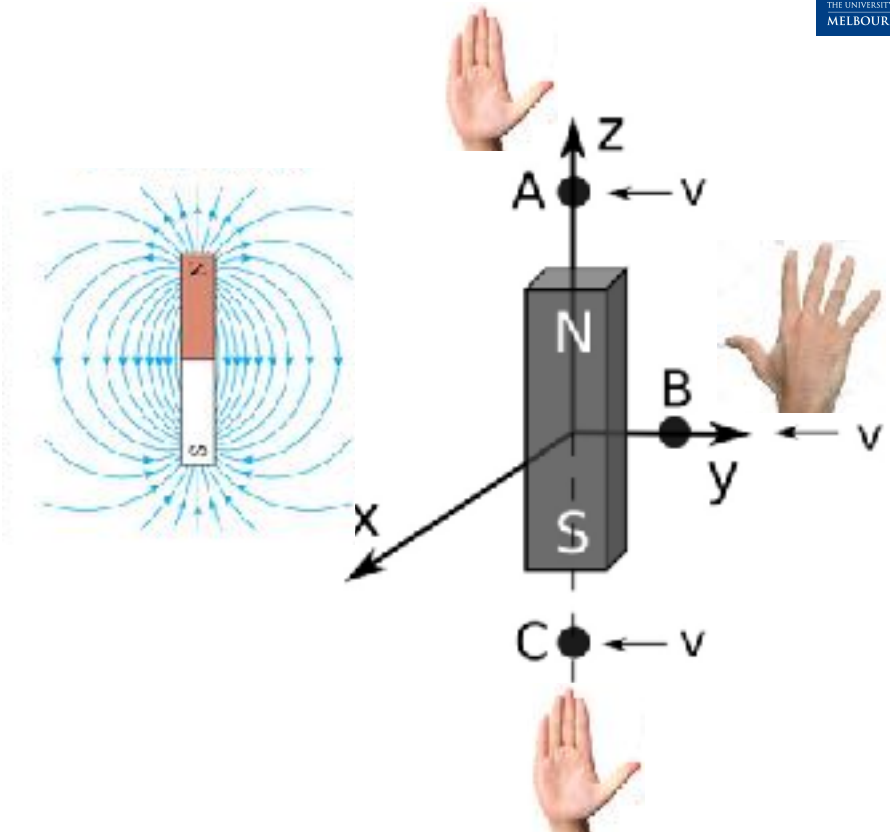
- (a) **A** $+\hat{x}$ **B** $-\hat{x}$ **C** $+\hat{x}$

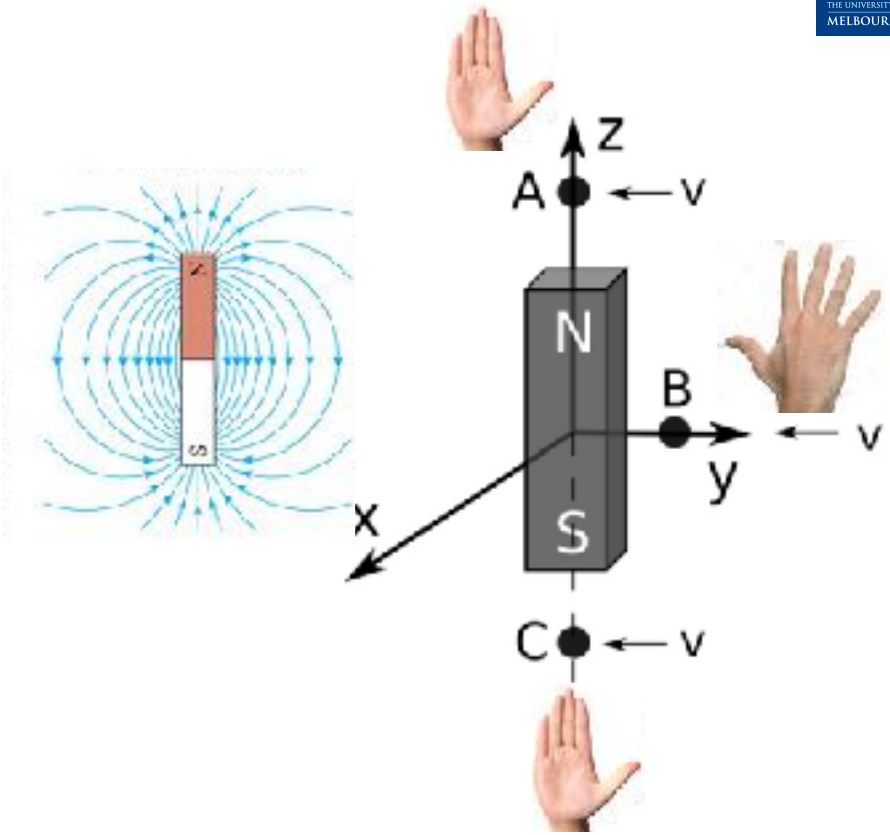


(b) Greater magnitude at A since the magnetic field is much stronger at the poles

(d)

2012 Exam Solutions

- (a) **A** $+\hat{x}$ **B** $-\hat{x}$ **C** $+\hat{x}$



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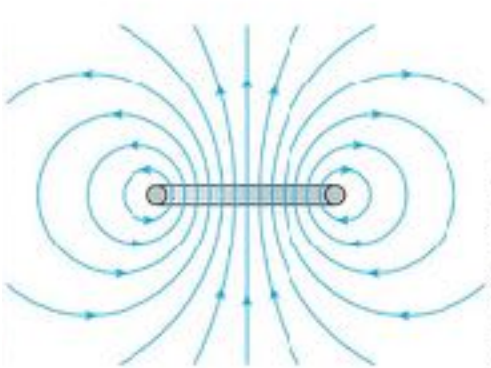
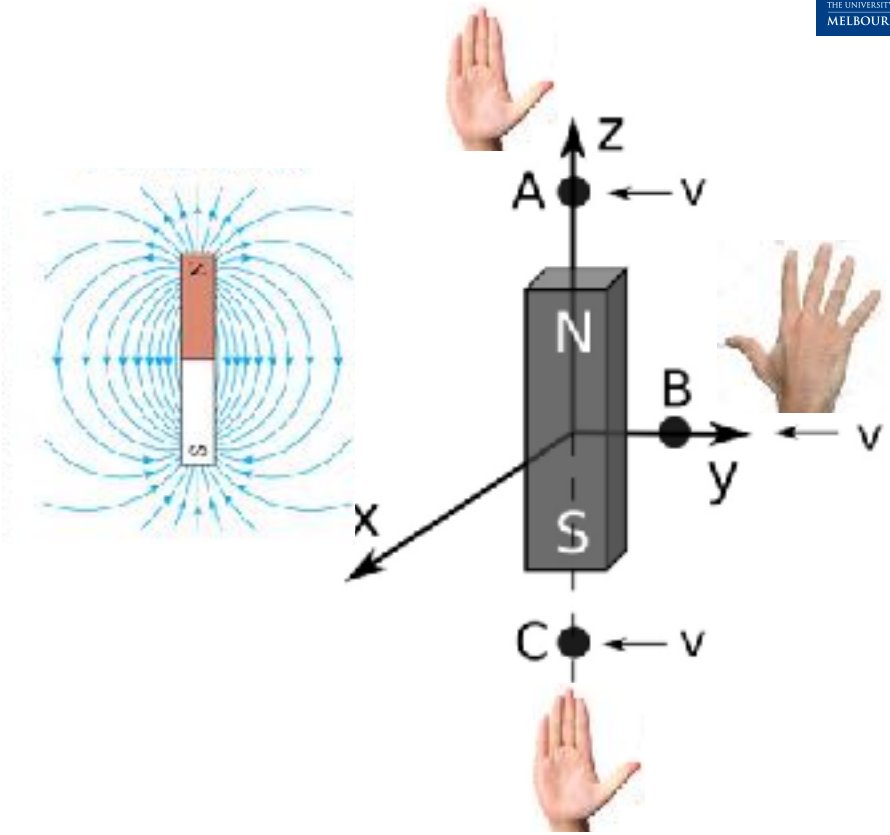


2012 Exam Solutions

- (a) **A** $+\hat{x}$ **B** $-\hat{x}$ **C** $+\hat{x}$



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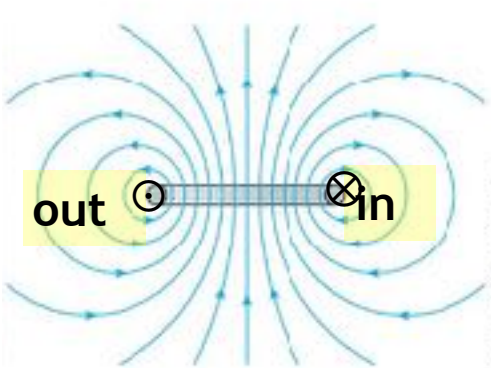
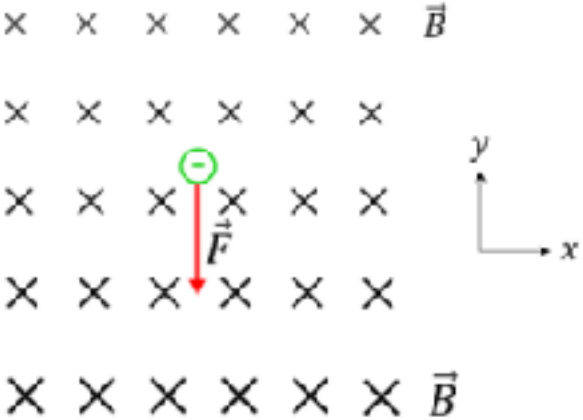


2015 exam

Question 8 **20 marks**

(a) A negatively charged particle is moving in a region of non-uniform magnetic field and experiences a force $\vec{F}$ in the direction $-y$ (see figure). The size of the × symbols is proportional to the magnetic field strength, such that $\vec{B}$ is greater where the symbols are larger.



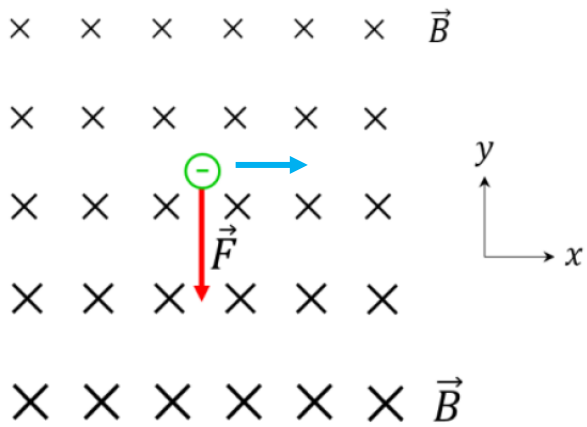
(i) What is the initial direction of the particle?

(ii) Does the magnitude of the force then increase, decrease, or remain the same?

(iii) Copy the diagram and sketch the trajectory of the particle.

2015 exam solutions

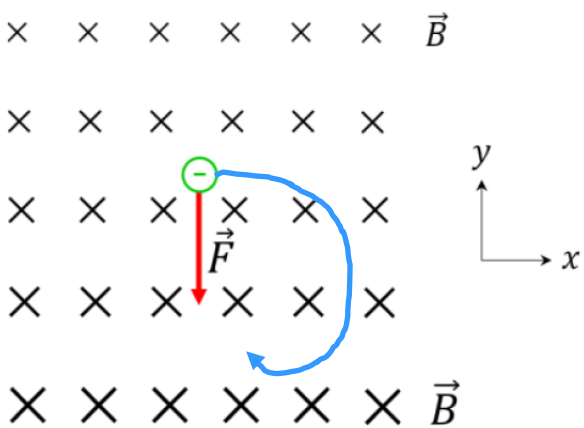
(a)



(i) fingers into page, palm down, thumb points in the $-\hat{x}$ direction. Charge is negative so velocity is $+\hat{x}$ direction.

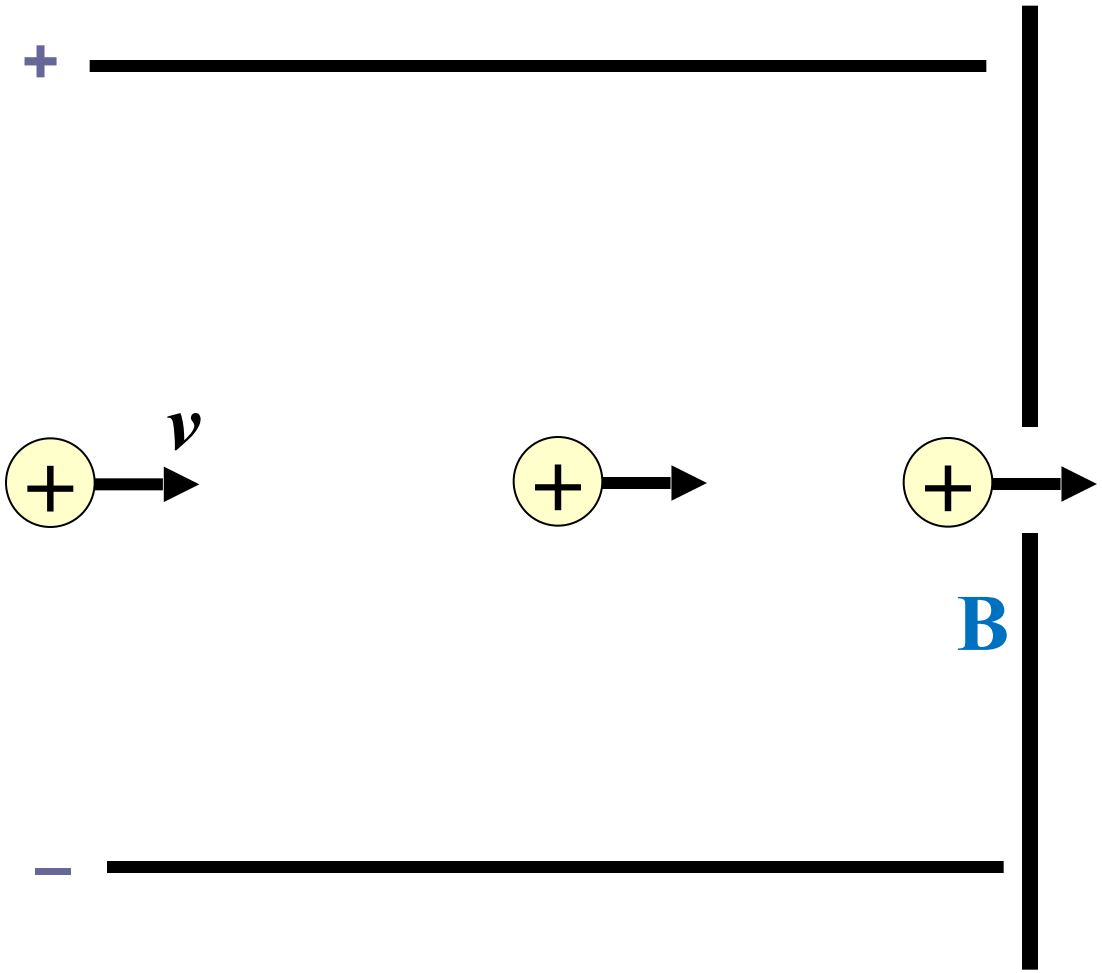
(ii) Force pulls the particle into a region of increasing magnetic field, speed is constant so force increases.

(iii)



Example: Wien velocity filter

- Crossed *E* and *B* filter ions by speed



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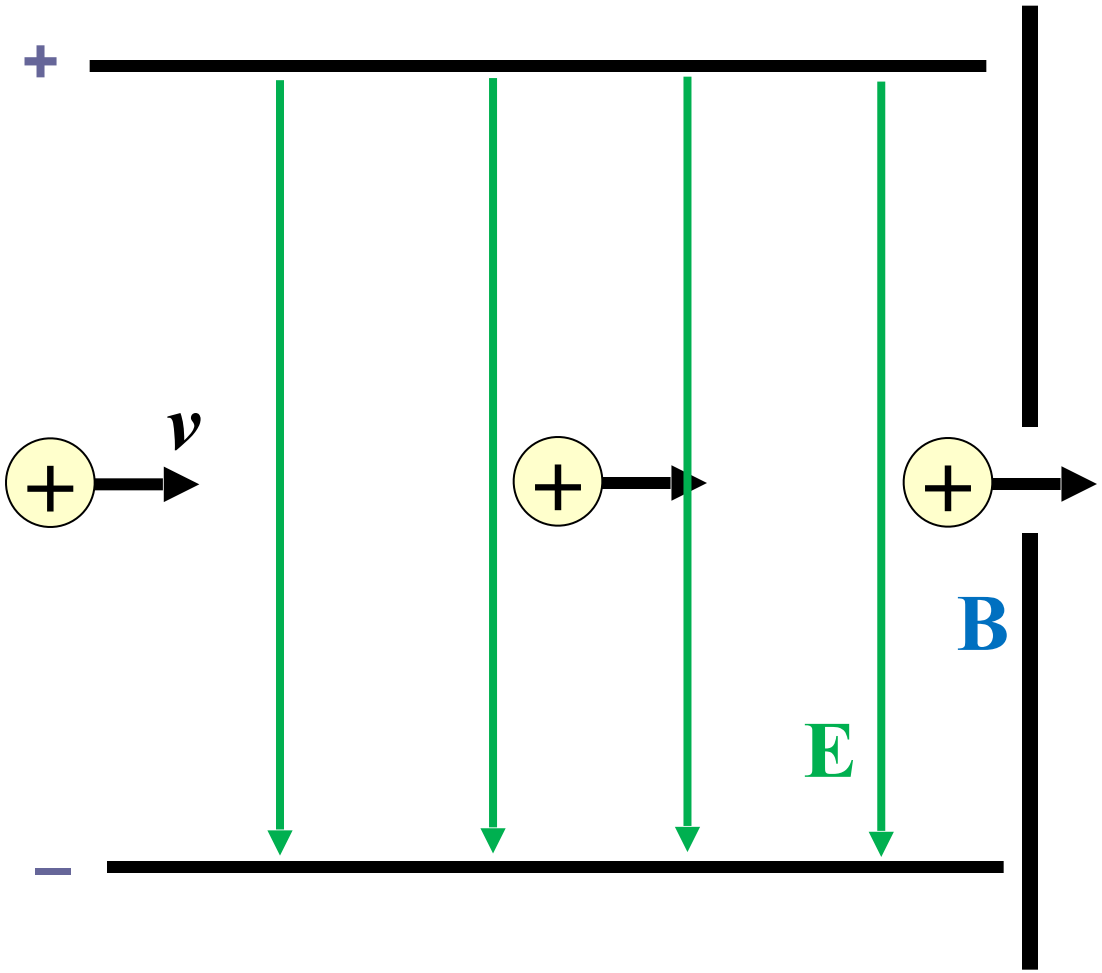

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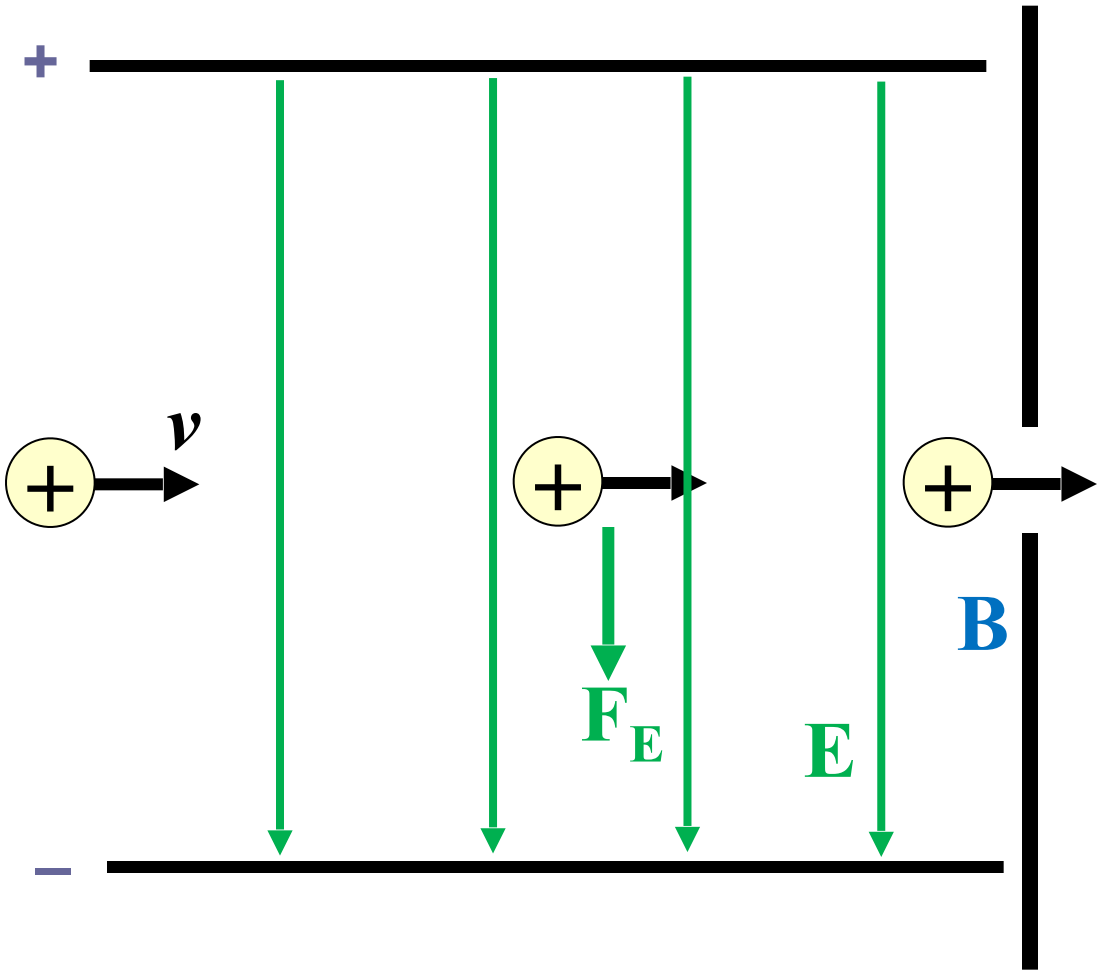

- Electric force: $F_E = qE$

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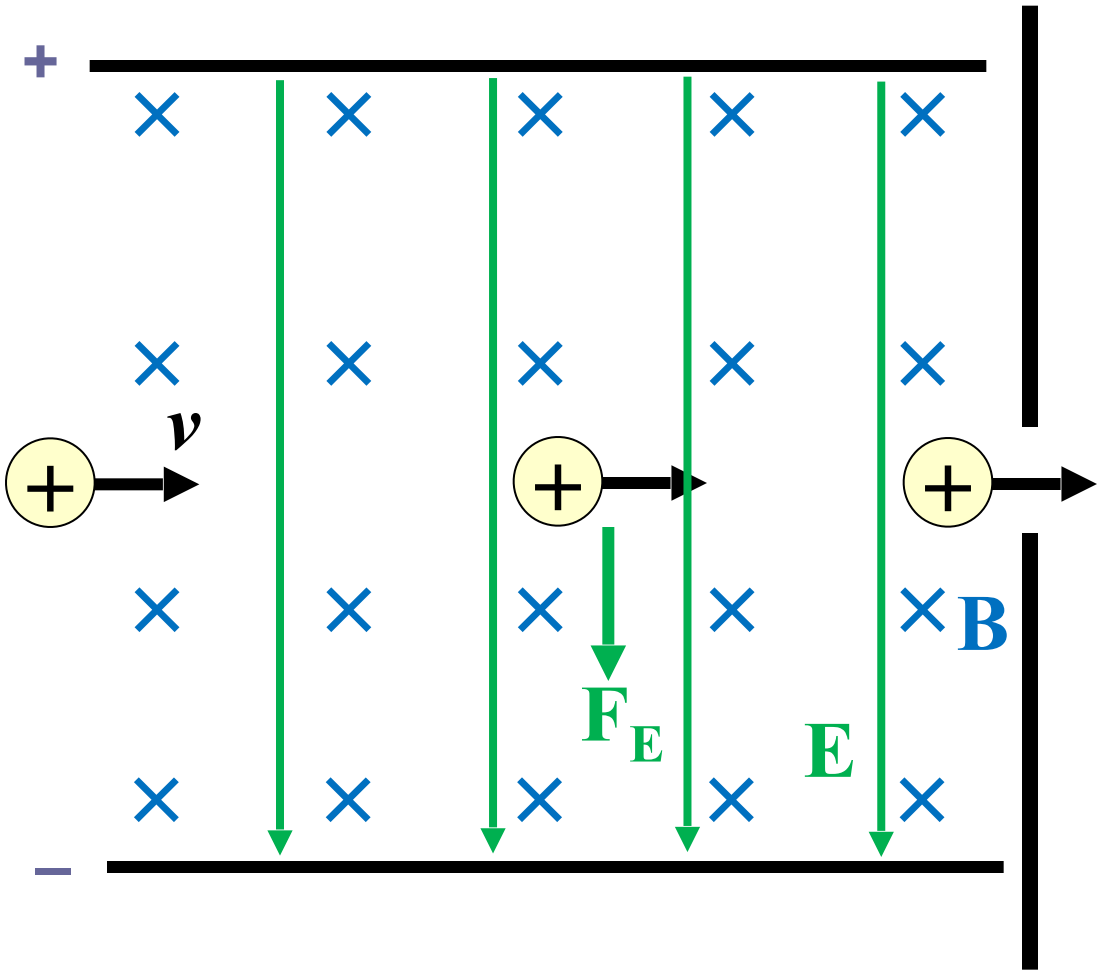

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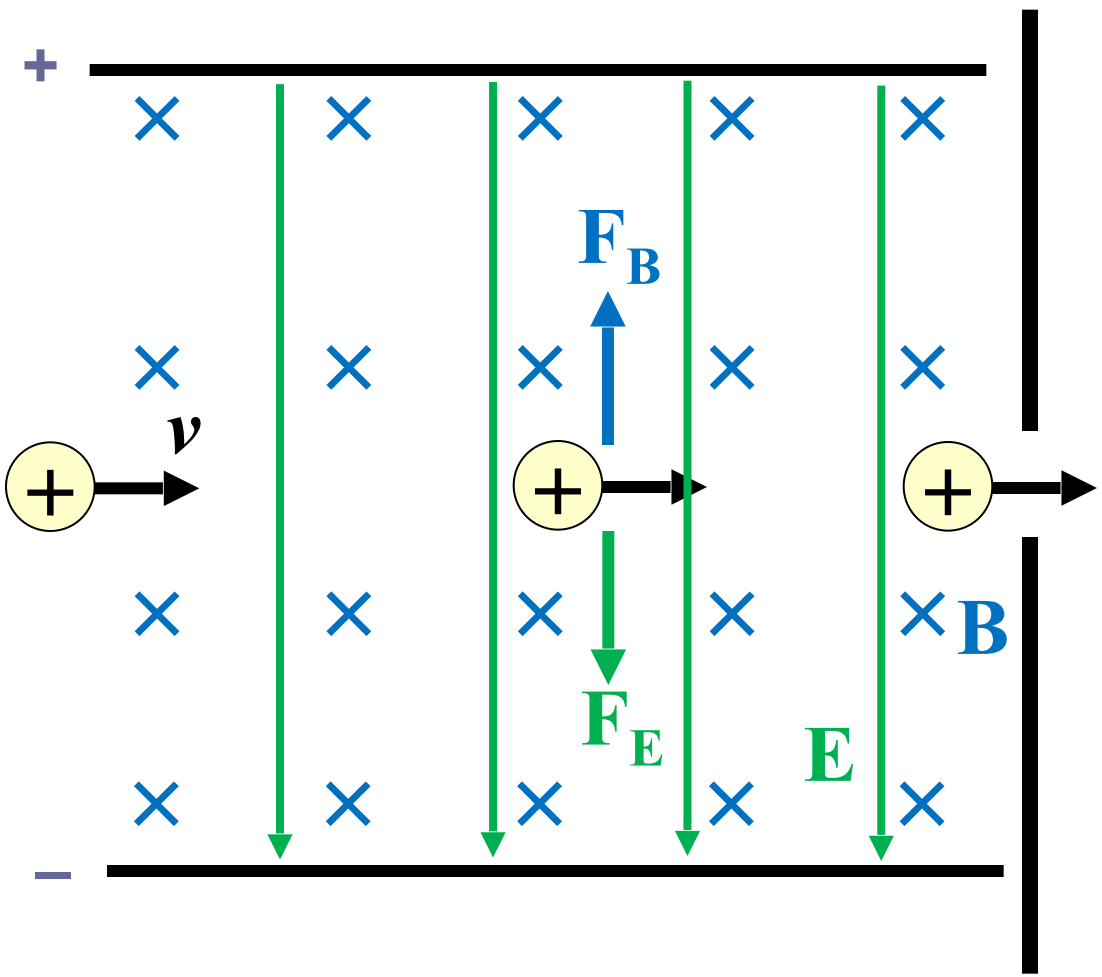

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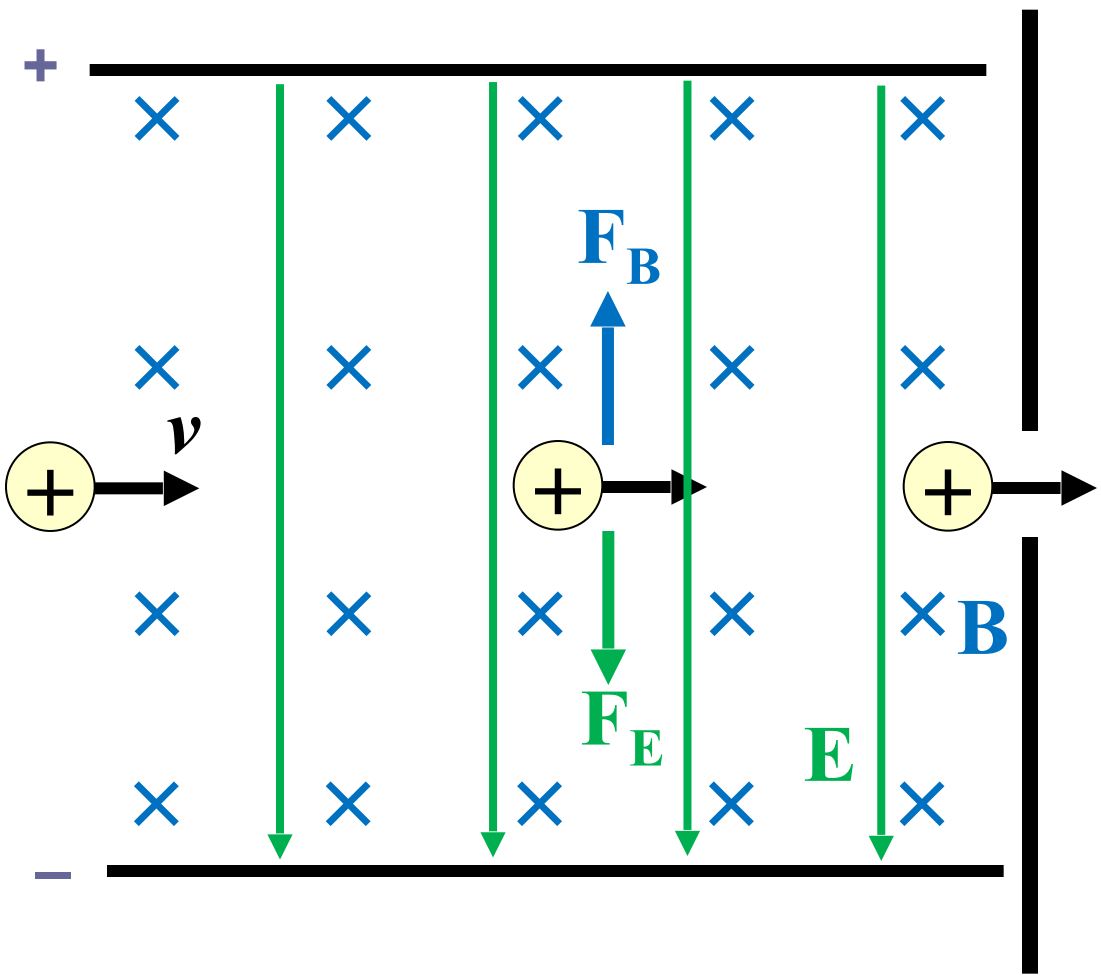
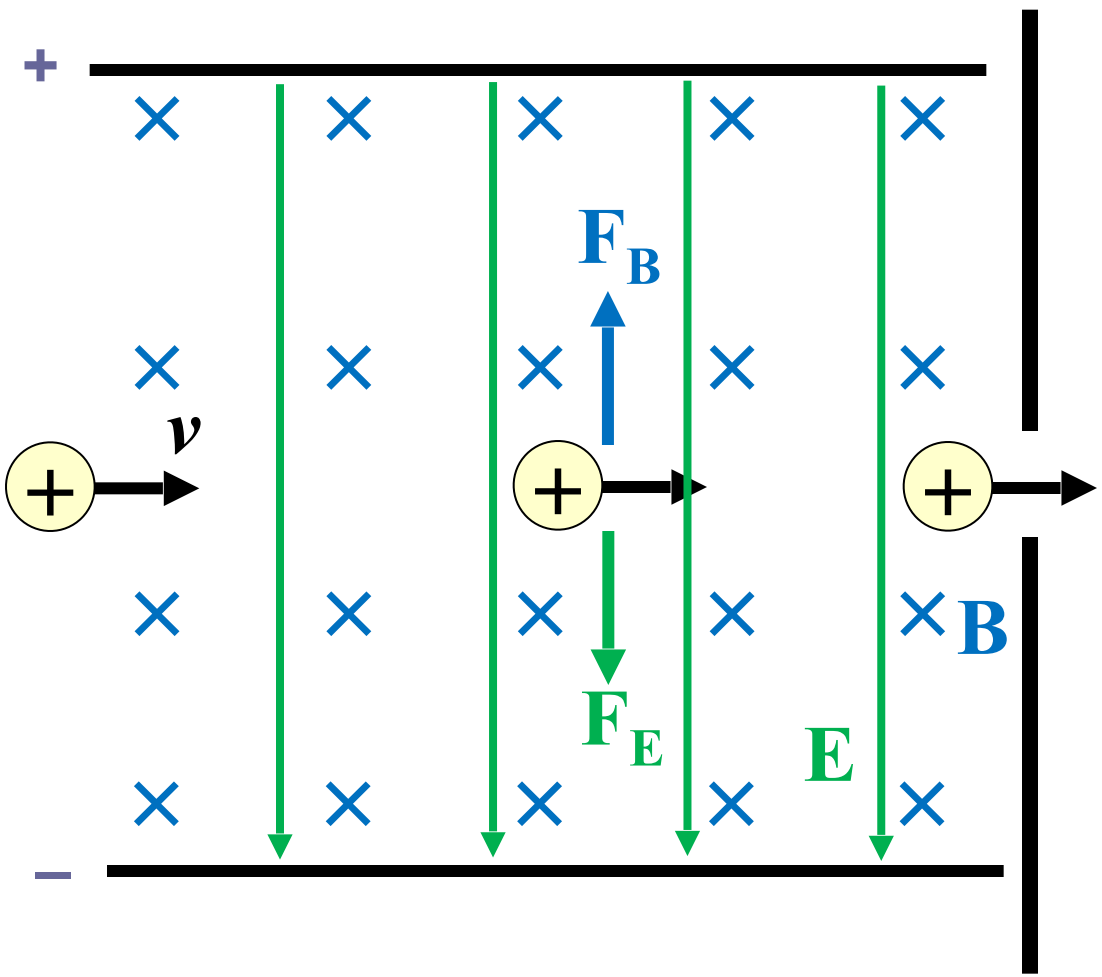


- Electric force: $F_E = qE$
- Magnetic force: $F_B = qvB$
- Adjust E for zero deflection so $F_E = F_B$
- Hence: $qE = qvB$

Wilhelm Wien 1898

Example: Wien velocity filter

- Crossed *E* and *B* filter ions by speed



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$$\boxed{v = E/B}$$

Wilhelm Wien 1898

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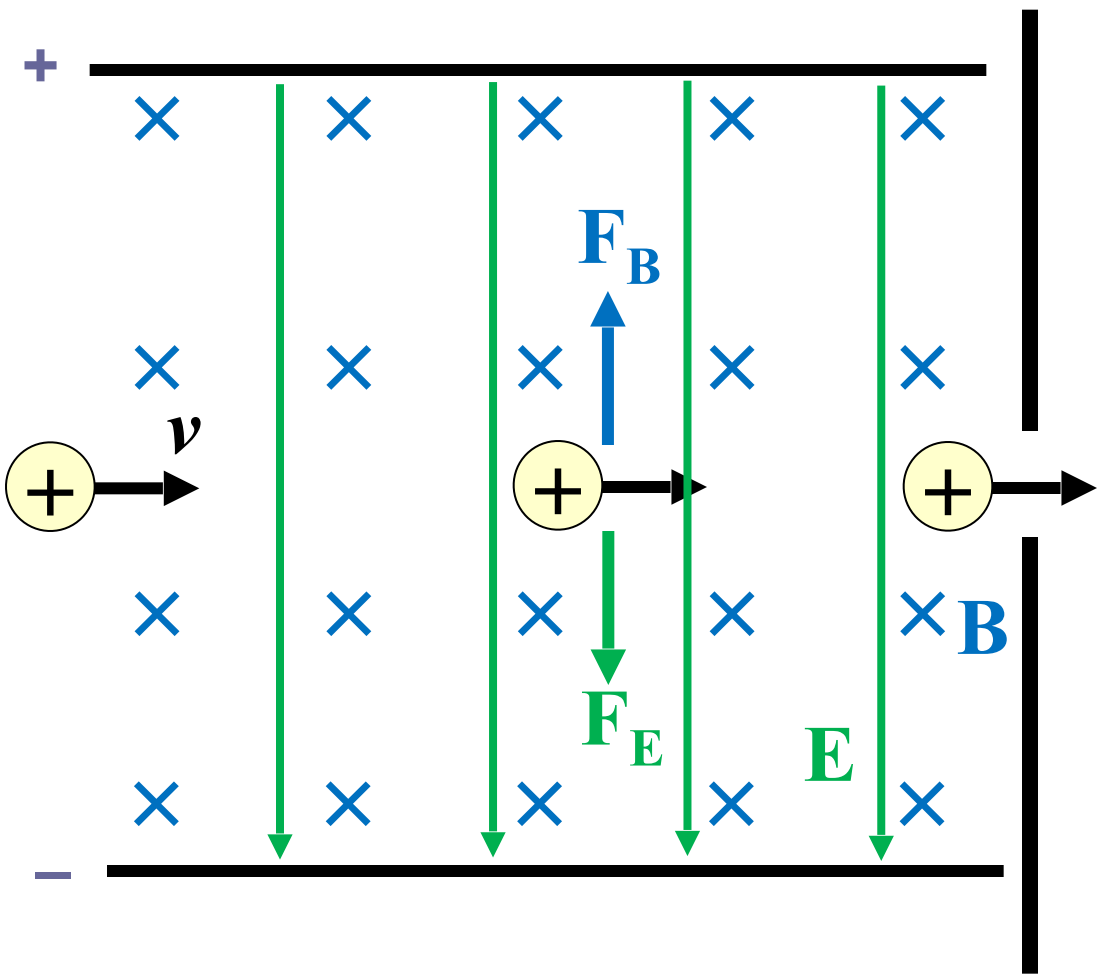
- Crossed *E* and *B* filter ions by speed



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$$\boxed{v = E/B}$$

- Too fast: $F_B > F_E$ bend up

Wilhelm Wien 1898

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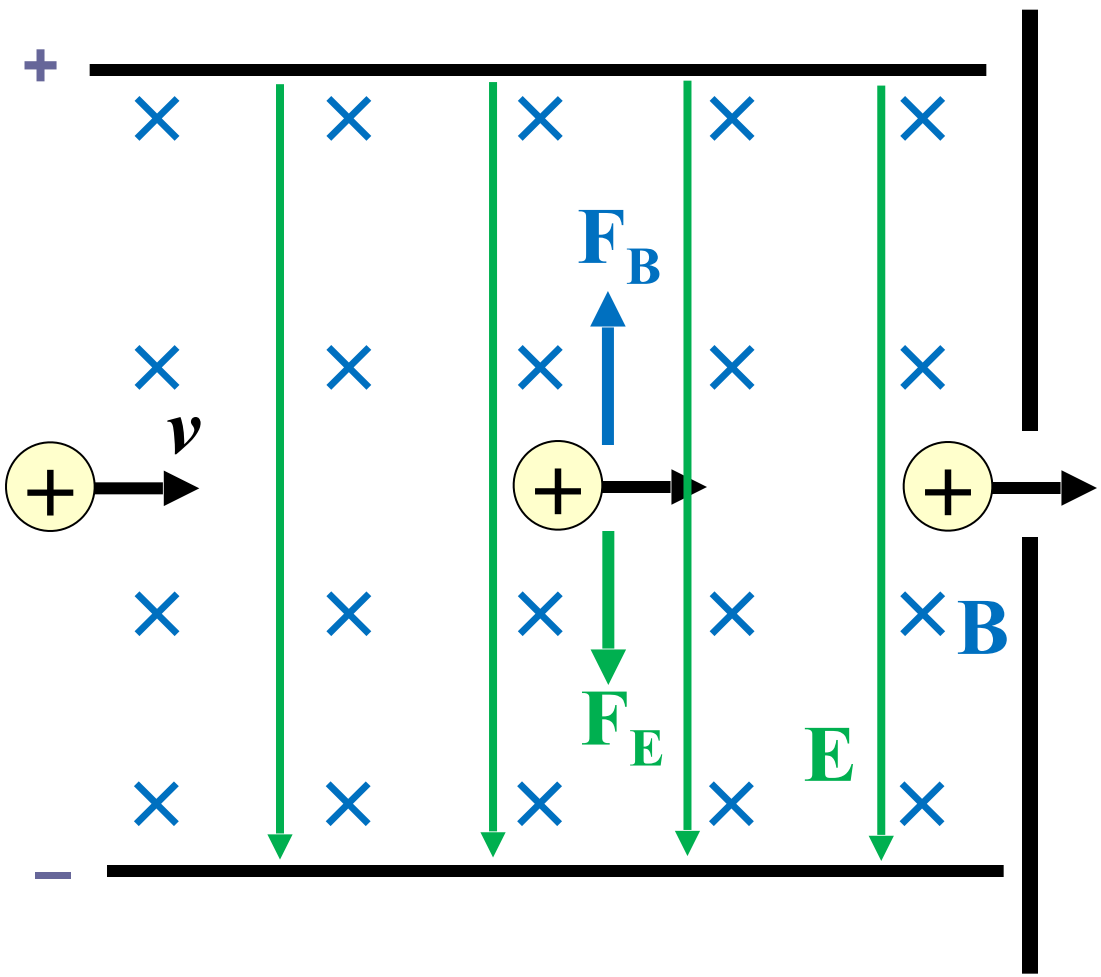


- Electric force: $F_E = qE$
- Magnetic force: $F_B = qvB$
- Adjust E for zero deflection so $F_E = F_B$
- Hence: $qE = qvB$

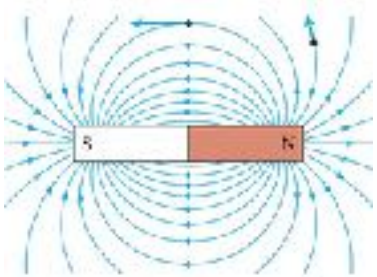
$$\boxed{v = E/B}$$

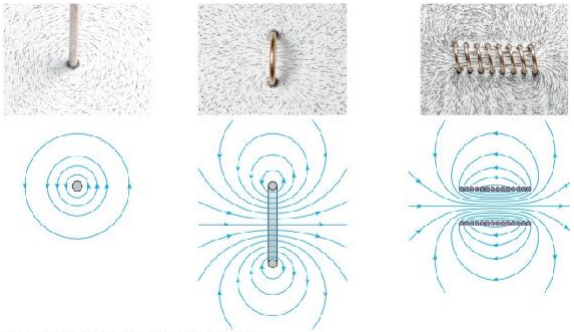
- Too fast: $F_B > F_E$ bend up
- Too slow: $F_B < F_E$ bend down

Wilhelm Wien 1898

Summary

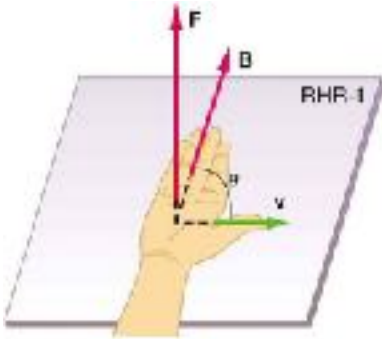
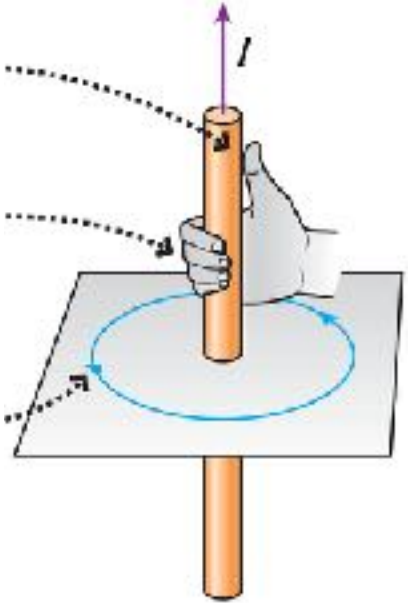
- Can measure magnetic field direction and strength with a compass needle
- Connecting directions gives a map of the magnetic field
- Electric currents generate magnetic fields: straight wire, loops, solenoids





$$B_{\text{wire}} = \frac{\mu_0 I}{2\pi r} \qquad B_{\text{loop}} = \frac{\mu_0 N I}{2r}$$

$$B_{\text{solenoid}} = \mu_0 I \frac{N}{\ell} = \mu_0 n I$$

- The force on a moving charge is

$$F = qvB\sin\theta$$